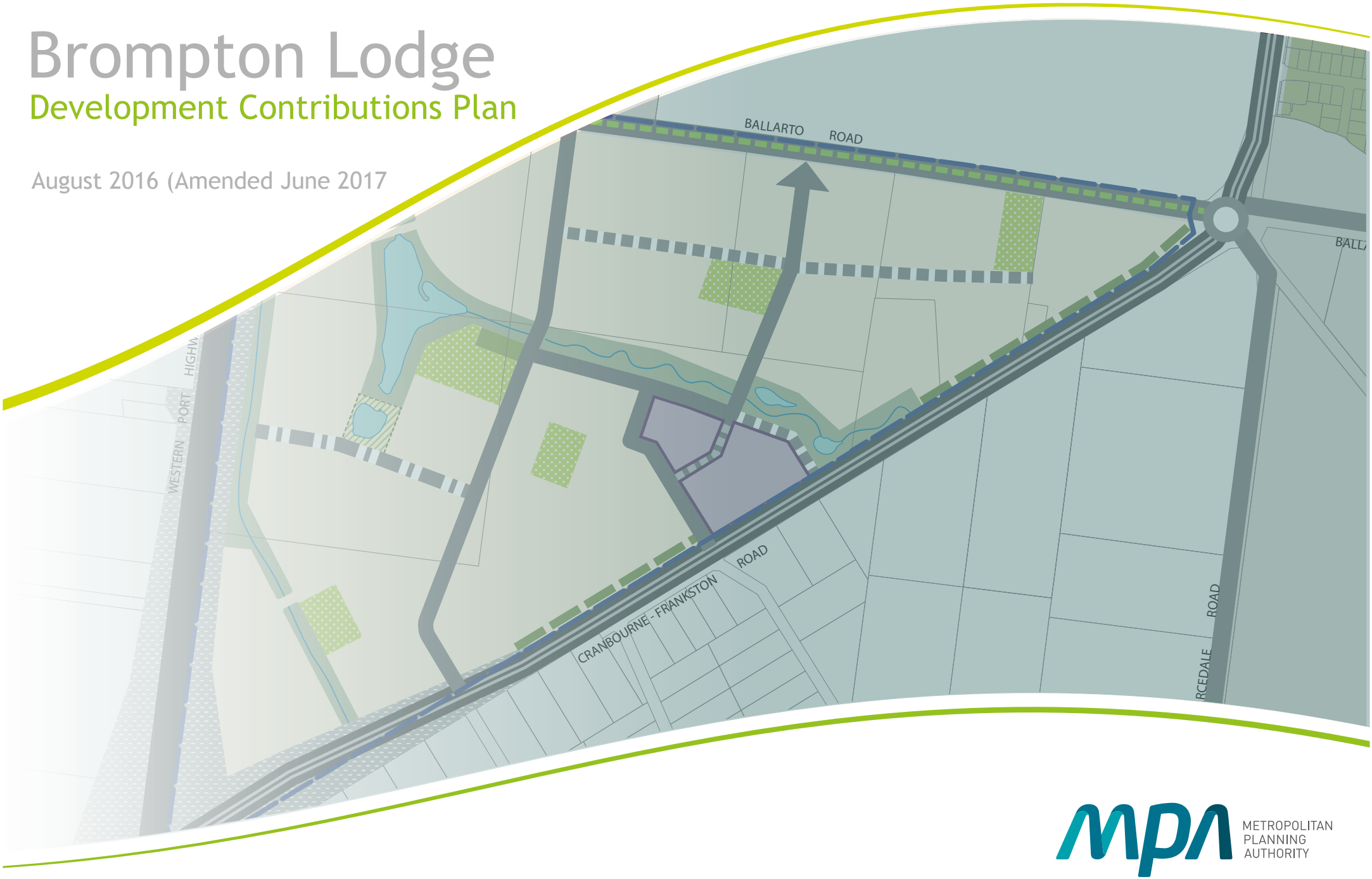
Brompton Lodge

Development Contributions Plan

August 2016 (Amended June 2017



METROPOLITAN PLANNING AUTHORITY

CONTENTS

1.0 INTRODUCTION 5

1.1 Report Structure 5
1.2 Strategic basis 7
1.3 Planning & Environment Act 1987 7
1.4 Strategic planning for Brompton Lodge and the Brompton Lodge PSP 7
1.5 Brompton Lodge Precinct Structure Plan 9
1.6 The area to which the Development Contributions Plan applies 10
1.7 Related infrastructure agreements 10
1.8 Project & property identification 10

2.0 INFRASTRUCTURE PROJECT JUSTIFICATION 11

2.1 Items not included in the Development Contributions Plan 11
2.2 Infrastructure Projects 13
2.3 Project Timing 16
2.4 Distinction between Development and Community Infrastructure 16

3.0 CALCULATION OF CONTRIBUTIONS 18

3.1 Calculation of Net Developable Area and Demand Units 18
3.2 Calculations of Contributions Charges 20

4.0 DEVELOPMENT CONTRIBUTIONS PLAN ADMINISTRATION 24

4.1 Payment of Contribution Levies and Payment Timing 24
4.2 Funds Administration 26
4.3 Construction and Land Value Costs Indexation 27
4.4 Development Contributions Plan Review Period 27
4.5 Collecting Agency (Agency Responsible for Collecting Infrastructure Levy) 27
4.6 Development Agency (Agency Responsible for Works) 27

5.0 IMPLEMENTATION STRATEGY 28

5.1 Rationale for the Implementation Strategy 28
5.2 Implementation Mechanism 28

6.0 OTHER INFORMATION 29

6.1 Acronyms 29
6.2 Glossary 29

7.0 APPENDICES 32

APPENDIX A Precinct Infrastructure Plan
APPENDIX B Property Specific Land Budget
APPENDIX C Interim Intersection Designs
APPENDIX D Interim Ballarto Road Design
APPENDIX E Ultimate Intersection designs
APPENDIX F Ultimate Ballarto Road Design
APPENDIX G Project Costings

TABLES

Amended by GC75

Table 1 Summary of Charges
Table 2 Charge Area
Table 3 Road and Intersection projects
Table 4 Local sports reserve and community facilities projects
Table 5 Community Infrastructure
Table 6 Summary Land Use Budget
Table 7 Infrastructure Project Timing and Calculation of Costs- Development Infrastructure Levy
Table 8 Infrastructure Project Timing and Calculation of Costs- Community Infrastructure Levy
Table 9 External Apportionment

PLANS

Plan 1 Future Urban Structure
Plan 2 Charge Area
Plan 3 Road and Intersection Projects
Plan 4 Local Sports Reserve and Community Facilities

Version	Date	Incorporated into the planning scheme by amendment	Description of changes
1	Dec 2016	C190	N/A
2	June 2017	GC75	Incorporate changes associated with Community Infrastructure Levy (CIL) increase.

The following table provides an overview of the project categories and charges included within this DCP. A more detailed explanation of apportionment, methods of calculation, and the description and costs of individual projects is included within the document.

Table 1 Summary of Charges

Amended by GC75

SUMMARY - NET DEVELOPABLE AREA (NDA) BY CHARGE AREA		
CHARGE AREA	TOTAL COST OF CONTRIBUTION	CONTRIBUTION PER NET DEVELOPABLE HECTARE (NDHA)
Residential (RNDA)	$26,019,543.81	$333,053

SUMMARY - DEVELOPMENT INFRASTRUCTURE LEVY (EXCLUDING EXTERNAL APPORTIONMENT)		
PROJECTS	TOTAL COST OF PROJECTS	CONTRIBUTION PER NET DEVELOPABLE HECTARE (NDHA)
Transport	$18,445,640	$236,106
Recreation	$3,791,069	$48,526
Community	$3,782,834	$48,421
Total	**$26,019,544**	**$333,053**

SUMMARY - BREAKDOWN OF DEVELOPMENT INFRASTRUCTURE LEVY		
PROJECTS	TOTAL COST OF PROJECTS	CONTRIBUTION PER NET DEVELOPABLE HECTARE (NDHA)
Land	$6,619,009	$84,724
Construction	$19,400,535	$248,329
Total	**$26,019,544**	**$333,053**

SUMMARY - COMMUNITY INFRASTRUCTURE LEVY		
	ESTIMATED DWELLINGS	ESTIMATED TOTAL CONTRIBUTION
Capped at $1,150 per dwelling	**1,484**	**$1,024,169**

1.0 INTRODUCTION

The Brompton Lodge Development Contributions Plan (the 'DCP') has been prepared by the Metropolitan Planning Authority with the assistance of City of Casey Council, service authorities and other major stakeholders.

The Brompton Lodge DCP (the DCP):

- Outlines projects required to ensure that future residents, visitors and workers in the area can be provided with timely access to infrastructure and services necessary to support a quality and affordable lifestyle.
- Establishes a framework for development proponents to make a financial contribution towards the cost of the identified infrastructure projects. It ensures that the cost of providing new infrastructure and services is shared equitably between various development proponents and the wider community.
- Provides the details of the calculation of financial contributions that must be made by future developments towards the nominated projects. In this way, it provides developers, investors and local communities with certainty about development contributions requirements and how these will be administered.

1.1 Report Structure

PART 1

Part 1 describes the strategic basis for the Development Contributions Plan.

PART 2

Part 2 sets out the justification for the various infrastructure projects included in the Development Contributions Plan.

PART 3

Part 3 sets out how the development contributions are calculated and costs apportioned.

PART 4

Part 4 focuses on administration and implementation of the Development Contributions Plan.

PART 5

Part 5 focuses on implementation of the Development Contributions Plan and identifies infrastructure delivery priorities.

PART 6 and 7

Parts 6 and 7 provide further information on projects, including cost estimate breakdowns, and appendices.


RANFURLIE GOLF CLUB
BALLARTO ROAD
BALLARTO ROAD
WESTERN PORT HIGHWAY
CRANBOURNE - FRANKSTON ROAD
CHEVRON AVE
WOODLANDS RD
PEARCEDALE ROAD
1:10,000 @ A4
v151019
0 100 200 300 400 500
N
Plan 1 - Future Urban Structure Plan
Brompton Lodge Development Contributions Plan
precinct boundary
residential
local town centre
existing urban area
freeway
arterial road (6 Lanes)
arterial road (4 Lanes)
connector street
main street - town centre
key local access street
local access street level 2 with 2 way bike path
waterway and drainage reserve
local park (within residential areas)
conservation area
Note: A tree reservation or frontage must be provided where residential lots interface with Cranbourne- Frankston Road and Western Port Highway.

1.2 Strategic basis

The strategic basis for the DCP is established by the State and Local Planning Policy Framework of the Casey Planning Scheme. The key documents are the:

- Municipal Strategic Statement.
- Brompton Lodge Precinct Structure Plan.
- Relevant Precinct Structure Plan supporting documents.

These documents set out a broad, long term vision for the sustainable development of the DCP area and its surrounds.

1.3 Planning & Environment Act 1987

The DCP has been prepared in accordance with Part 3B of the Planning and Environment Act 1987 (the Act) as well as other relevant legislation and has been developed in line with the State and Local Planning Policy Framework of the Casey Planning Scheme. It is consistent with the Minister for Planning's Direction on Development Contributions made under section 46M(1) of the Act and has had regards to the Victorian Government's Development Contributions Guidelines (the 'DCP Guidelines').

The DCP provides for the charging of a Development Infrastructure Levy pursuant to section 46J(a) of the Act towards works, services and facilities. It also provides for the charging of a Community Infrastructure Levy pursuant to section 46J(b) of the Act as some items are classified as community infrastructure by reference to the Act, the Minister's Direction on Development Contributions and the DCP guidelines.

The DCP forms part of the Casey Planning Scheme pursuant to section 46I of the Act and is an incorporated document under Clause 81 of the Casey Planning Scheme.

The Development Contributions Plan Overlay applies to the area shown in Plan 1.

1.4 Strategic planning for Brompton Lodge and the Brompton Lodge PSP

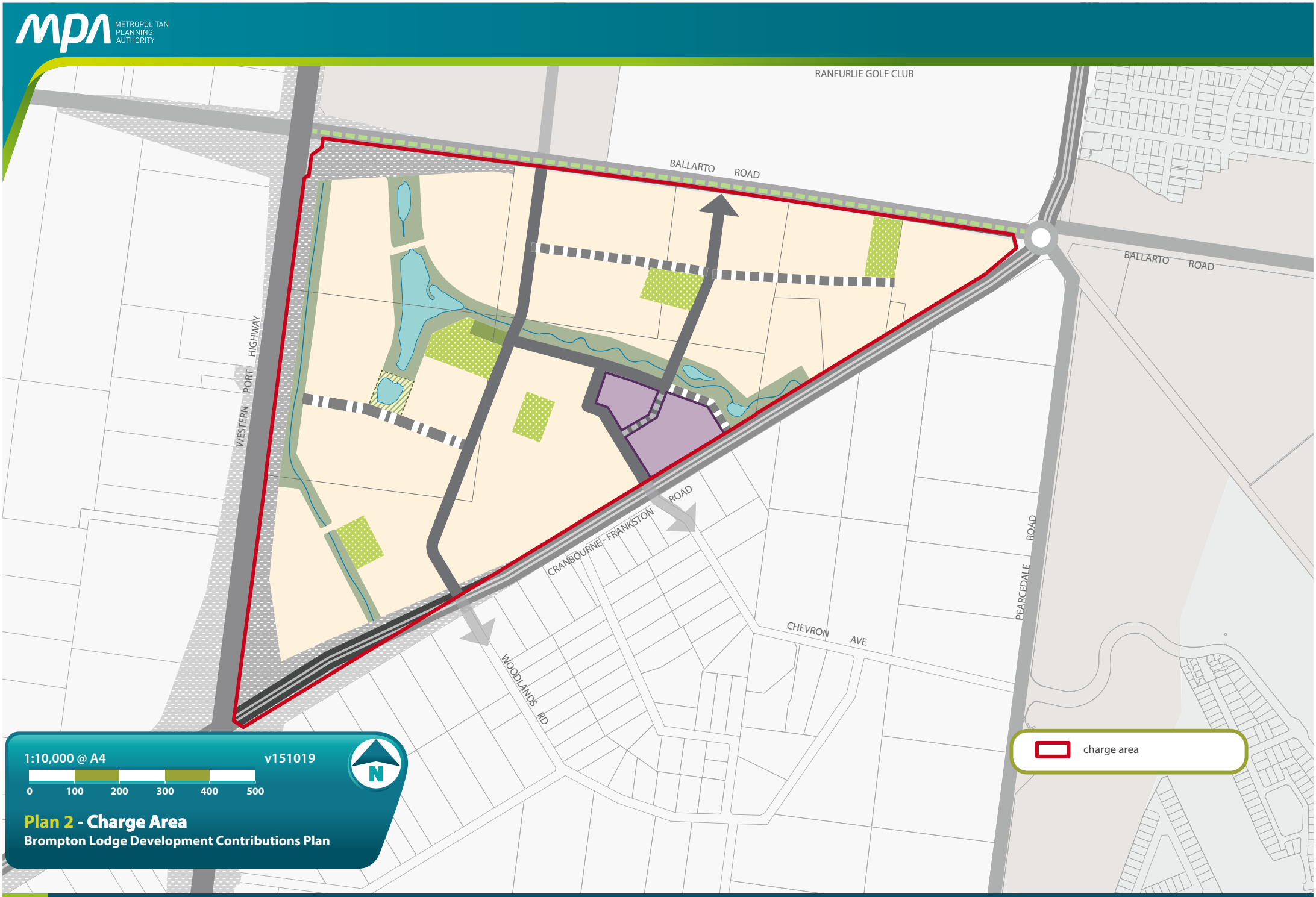
The area of Brompton Lodge covered by the DCP consists of 107 hectares of land in the Urban Growth Zone.

Planning for the Brompton Lodge precinct has been undertaken in consideration of the Cranbourne West precinct and surrounding areas. This is particularly important for the Brompton Lodge precinct given that its residents will utilise the schools, community centres and active open space outside of the precinct.

The need for the infrastructure included within the DCP has been determined according to the anticipated development of the Brompton Lodge area.

The DCP has been prepared in conjunction with the Brompton Lodge Precinct Structure Plan as it provides the rationale and justification for infrastructure items that have been included. Accordingly, the DCP is an implementation based planning tool which identifies the infrastructure items required by the new community and apportions the cost of this infrastructure in an equitable manner across the plan area.

The Background Reports for the precinct provide an overview of the planning process for the Brompton Lodge area.


METROPOLITAN PLANNING AUTHORITY
RANFURLIE GOLF CLUB
BALLARTO ROAD
BALLARTO ROAD
WESTERN PORT HIGHWAY
CRANBOURNE-FRANKSTON ROAD
CHEVRON AVE
WOODLANDS RD
PEARCEDALE ROAD
charge area
1:10,000 @ A4
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0 100 200 300 400 500
N
Plan 2 - Charge Area
Brompton Lodge Development Contributions Plan

1.5 Brompton Lodge Precinct Structure Plan

The Brompton Lodge PSP (the PSP) sets out the vision for how land should be developed, illustrates the future urban structure and describes the objectives to be achieved by the future development. It also outlines projects required to ensure that future residents, visitors and workers within the area are provided with timely access to services and transport infrastructure necessary to support a quality affordable lifestyle.

The PSP enables urban development and the future urban structure of the new community and connections through to the wider area is depicted through the transport network, open space, town centre and housing.

The PSP will ultimately accommodate approximately 4,156 people and 1,484 dwellings. It is also proposed to deliver a local town centre with approximately 6,280m^2 of retail floorspace with the potential to deliver 369 jobs.

The wider Brompton Lodge area planning process confirmed:

- A detailed NDA based on a survey of encumbrances.
- All road and intersection projects required to service the new community.
- The contribution towards sporting reserves and community centres.

Accordingly, the PSP may refine the location and layout of some items included in this DCP but it will not change the project list nor the overall rate.

1.6 The area to which the Development Contributions Plan applies

In accordance with section 46K(1)(a) of the Planning and Environment Act 1987 the DCP applies to land shown in Plan 1. The area is also shown on Development Contributions Plan Overlay Schedule 19 in the Casey Planning Scheme.

Table 2 Charge Area

	AREA	LABEL
Area 1	78.12	Residential

The DCP applies to approximately 78.13 Net Developable Hectares of land and is defined as one charge area. The DCP clearly demonstrates the infrastructure required to service urban development in the Brompton Lodge PSP.

In selecting items, consideration has been given to ensure they are not already wholly funded through another contribution mechanism, such as a mandatory infrastructure construction requirement of the Brompton Lodge PSP, an existing local development contributions plan, an agreement under Section 173 of the Act or as a condition on an existing planning permit. Identified overlap in funding has been addressed, for example by adjusting other relevant development contributions plans or other suitable means provided for in the Planning and Environment Act.

1.7 Related infrastructure agreements

Some infrastructure items are apportioned across the adjoining DCP area of Cranbourne West.

The apportionments listed in Table 7 and 8 have taken this into consideration. Refer to Table 9 for details of external apportionment.

1.8 Project & property identification

1.8.1 Project identification

The project identification system used in the DCP has been designed to assist in understanding of and navigation through the document. Road, bridge, intersection, and community facility projects use the identification system of project category, PSP number, and a sequential project number. As an example, a road project will have the project identifier similar to RD01. The table below provides a summary of the project categories.

PROJECT CATEGORIES	
RD	Roads
IN	Intersections
CO	Community Facilities
LS	Local Sports Facilities
P	Pavillion

Sporting reserves are given different identifiers relating to the rates applied to broad projects. LS denotes land and construction costs and P denotes pavilions.

2.0 INFRASTRUCTURE PROJECT JUSTIFICATION

The need for infrastructure included in the DCP has been determined according to the anticipated development of the Brompton Lodge area.

Items can be included in a development contributions plan if they will be used by the future community of an area. New development does not have to trigger the need for new items in its own right. The development is charged in line with its projected share of use. An item can be included in a development contributions plan regardless of whether it is within or outside the DCP area.

Before inclusion in the DCP, all items have been assessed to ensure they have a relationship or nexus to proposed development in the DCP area. The cost apportionment methodology adopted in the DCP relies on the nexus principle. A new development is deemed to have a nexus with an item if its future residents or employees are expected to make use of that item.

A summary of how each item is related to proposed development within the DCP area is set out below and individual item apportionments are identified in Table 8.

The items that have been included in the DCP all have the following characteristics:

- They are essential to the health, safety and well being of the community.
- They will be used by a broad cross-section of the community.
- They reflect the vision and objectives expressed in the Brompton Lodge PSP.
- They are not recurrent items.
- They are the basis for the future development of an integrated network.

2.1 Items not included in the Development Contributions Plan

The following items are not included in the DCP, as they are considered to be normal to the construction of a development and are not considered to warrant cost sharing arrangements beyond those set out in this DCP and must be provided by developers as a matter of course and in implementing the PSP:

- Internal streets and connector streets, creek, drainage line and utility easement crossings and associated traffic management measures (except where nominated in this DCP).
- Waterway management works and drainage systems.
- Intersections connecting the development to the existing road network, except where specified as development contributions plan projects.
- Water, sewerage, underground power, gas and telecommunications services.
- Local pathways and connections to the regional and / or district pathway network.
- Shared path networks, including required crossings of waterways.
- Basic levelling, provision of biodiversity offsets, water tapping and landscaping of local active and passive open space.
- Passive public open space reserve master plans and any agreed associated works required by the PSP.
- Council's plan checking and supervision.
- Bus stops.

These items may be further addressed and defined by an agreement under s173 of the Act and / or conditions in planning permits.

Construction of the following items has not been included within the DCP as they are determined to be State Infrastructure items:

- Subsequent carriageways of
  - Cranbourne-Frankston Road
  - Westernport Highway
- Government primary or secondary schools.

The delivery of the State Infrastructure items will be provided as warranted and as funds become available, which may potentially include Growth Area Infrastructure Charge (GAIC) Work in Kind projects.

Land for the following items has not been included within the DCP:

- Existing areas affected by the Public Acquisition Overlay in the Casey Planning Scheme for Westernport Highway.

If any roads not considered a State responsibility require duplication beyond that allowed for in this DCP, this would be undertaken by Council.



Plan 3 - Road and Intersection Projects
Brompton Lodge Development Contributions Plan

2.2 Infrastructure Projects

The following three types of projects are included in the DCP (refer to Plans 3 and 4):

- Transport
- Recreation
- Community

2.2.1 Transport projects

The transport related projects in the DCP are based on the transport network depicted in Plan 3 which is supported by the Brompton Lodge PSP. The transport projects include a combination of:

- Road construction (including land requirements)
- Construction of major controlled intersections and associated works

The following and intersection road projects shown in Plan 2 are funded by the DCP:

Table 3 Road and Intersection Projects

DCP PROJECT ID	PROJECT TITLE PROJECT DESCRIPTION	CHARGE AREAS CONTRIBUTING	HECTARES CONTRIBUTING	INDICATIVE PROVISION TRIGGER
RD-01	Ballarto Road widening Purchase of land to widen road reserve (ultimate treatment) including land for intersections and construction of road (interim treatment) from Western Port Hwy to Cranbourne-Frankston Rd.	Residential	78.12	At time of subdivision
RD-02	Chevron Avenue upgrade Upgrade of existing carriageway for 587m, excluding intersections, to an urban standard. Shared path for entire length	Residential	78.12	At time of subdivision
IN-01	Ballarto Road and Woodlands Road Purchase of land for intersection (ultimate treatment) and construction of roundabout (interim) and Pedestrian signals	Residential	78.12	At time of subdivision
IN-02	Ballarto Road and Eastern connector Purchase of land for intersection (ultimate treatment)	Residential	78.12	At time of subdivision
IN-03	Cranbourne-Frankston Road and Woodlands Road Purchase of land for intersection (ultimate treatment) and construction of arterial to connector signalised 4-way intersection (interim treatment)	Residential	78.12	At time of subdivision
IN-04	Cranbourne-Frankston Road and Eastern connector Purchase of land for intersection (ultimate treatment) and construction of arterial to connector signalised 4-way intersection (interim treatment)	Residential	78.12	At time of subdivision
IN-05	Ballarto Road/Cranbourne-Frankston Road/Pearcedale Road Construction of 5th leg of arterial to connector intersection (interim treatment)	Residential	78.12	At time of subdivision
IN-06	Western Port Hwy and Ballarto Road completion of fourth leg of Highway to connector roundabout (interim treatment)	Residential	78.12	At time of subdivision



Plan 4 - Local Sport Reserves and Community Facilities

Brompton Lodge Development Contributions Plan

2.2.2 Recreation facility project

The recreation contributions include:

- Provision of land for a sports reserve external to the precinct.
- Contribution towards construction of a sports facility in an open space reserve.
- Contribution towards construction of pavilions for future sports fields.

The DCP provides a contribution for a local sports reserve for the basic construction, surfacing, and landscaping of facilities.

2.2.3 Community facility projects

The community projects include a contribution towards:

- Land and construction of a Level 1 community centre within the Cranbourne West PSP area.
- Upgrade of a Level 2 community centre within the Cranbourne West PSP area.

The Brompton Lodge PSP is providing a contribution towards the construction of a new facility as well as the upgrade of a proposed community facility within the Cranbourne West PSP area. The community facilities will service a wider area than the Brompton Lodge PSP. The contribution towards these projects by this DCP reflect the best estimates of the future requirements and specifications of the future Brompton Lodge community. The community project contributions in this DCP are consistent with the range of facilities and provision ratio set out in research undertaken by the Metropolitan Planning Authority and ASR Research.

The detailed design and scope (as defined in Table 4) of each of the following community projects will be reviewed by the Development Agency closer to the time that they are constructed.

In reviewing the scope of the facility, the Development Agency will have regard to matters such as changing provision standards and models, the immediate needs of the community, current regulations and best practice and may adjust and refine the scope of the facility to respond to these matters.

The Development Agency may also adjust and refine the scope of the facility to reflect the capacity of any non-DCP funds that can be made available (for example from a grant or some other funding stream).

In adjusting and refining any final project scope the Development Agency will ensure that at least the same total cost of the project item (as indexed from time to time) is invested into the community facilities proposed.

The recreation + community projects shown in Plan 4 are funded by the DCP:

Table 4 Local Sports Field Project and Community Facilities

DCP PROJECT ID	PROJECT TITLE PROJECT DESCRIPTION	CHARGE AREAS CONTRIBUTING	HECTARES CONTRIBUTING	INDICATIVE PROVISION TRIGGER
LS-01	Local sports reserve at 1070 Cranbourne-Frankston Road, Cranbourne. Contribution towards land and construction of a local sports reserve at 1070 Cranbourne-Frankston Road, Cranbourne including a football/cricket oval and 6 tennis courts	Residential	78.12	At time of subdivision
CO-01	Family and Childrens Centre- 1/630 Hall Road Contribution towards the purchase of land and construction of Family and Children's Centre (Level 1) at 1/630 Hall Road, Cranbourne West to be constructed with 3 kindergarten rooms, 1 multi-purpose room, 1 meeting room and 2 maternal child health care rooms.	Residential	78.12	At time of subdivision
CO-02	Community Centre- Cranbourne West Activity Centre Contribution towards the upgrade of the future integrated community centre in the Cranbourne West Activity Centre as a Level 2 facility to include Neighbourhood House Community Learning Centre (NHCLC) spaces, halls, meeting rooms; arts and social support spaces, Community Service Organisation (CSO) spaces and a business accelerator (100m2) for Economic Development.	Residential	78.12	At time of subdivision

2.3 Project Timing

Each item in the DCP has an assumed indicative provision trigger specified in tables 3 to 6. The timing of the provision of the items is consistent with information available at the time that the DCP was prepared. The Council, as Development Agency will monitor and assess the required timing for individual items and have regard to its capital works program, the staging of the PSP and areas external to the DCP Area and the indicative provision trigger within Table 8. The Council as Development Agency will monitor and assess the required timing for individual items and have regard to its capital works program, the staging of the PSP's and areas external to the DCP area.

The Collecting Agency may consider alternatives to the priority delivery of works or land where:

- Infrastructure is to be constructed / provided by development proponents as works or land in kind, as agreed by the Collecting Agency.
- Network priorities require the delivery of works or land to facilitate broader road network connections.
- Community needs determine the delivery of works or land for community facilities or active open space.

All items in this DCP will be provided as soon as is practicable and as soon as sufficient contributions are available, consistent with Section 4.1 and acknowledging the Development Agency's capacities to provide the balance of funds not recovered by this DCP.

All items included in the DCP will be provided within 30 years from the date upon which this DCP was first incorporated in to the Casey Planning Scheme.

2.4 Distinction between Development and Community Infrastructure

In accordance with the Planning and Environment Act 1987 and the Ministerial Direction on Development Contributions, the DCP makes a distinction between 'development' and 'community' infrastructure.

The timing of payment of contributions is linked to the type of infrastructure in question.

The community infrastructure levy contributions are made by the home builder at the time of building approval (unless an alternative time is agreed between the collecting agency and a development proponent). Community infrastructure levy contributions will be paid for at a 'per-dwelling' rate. The Planning and Environment Act 1987 currently stipulates that the amount that may be contributed under a community infrastructure levy is no more than $1,150 per dwelling.

Amended by GC75
Deleted by GC75
Inserted by GC75

Variation to the Community Infrastructure Levy (CIL) payable under this Development Contributions Plan (DCP)

If the maximum amount of the CIL which may be collected under an approved DCP is varied under section 46L of the Planning and Environment Act 1987, the collecting agency may adjust the amount of the CIL payable under this DCP in accordance with section 46L of the Planning and Environment Act 1987.

The collecting agency must publish the adjusted amount of the CIL payable under this DCP on its website.

The projects listed in Table 5 are deemed to be community infrastructure levy projects.

All other infrastructure projects are considered to be development infrastructure projects.

Contributions relating to development infrastructure are to be made by developers at the time of subdivision. If subdivision is not applicable payments must be made prior to construction works.

Table 5 Community Infrastructure Levy Projects

DCP PROJECT ID	PROJECT TITLE PROJECT DESCRIPTION
P-01	Contribution towards construction of pavillions at 1070 Cranbourne-Frankston Road, Cranbourne.

3.0 CALCULATION OF CONTRIBUTIONS

The general cost apportionment method includes the following steps:

- Calculation of the Net Developable Area ('NDA') and demand units (refer Table 6).
- Calculation of project costs (refer Tables 7 and 8).
- Identification and allowance for external apportionment (refer Table 9).
- Cost apportionment and catchments (refer Table 1).
- Identification of development types required to pay the levy (refer Table 2).
- Calculation of costs payable for each infrastructure project (refer Table 7 and 8).
- The development infrastructure charge per hectare for each development type and the community infrastructure levy per dwelling (refer Table 7 and 8).

3.1 Calculation of Net Developable Area and Demand Units

The following section sets out how NDA is calculated and outlines the development projections anticipated for the area. Calculations of NDA for each individual property is outlined in the property-specific land budget included within the relevant PSP.

3.1.1 Net Developable Area

In this DCP, all development infrastructure contributions are payable on the net developable area of land on any given development site.

For the purposes of this DCP the NDA is defined as the total amount of land within the precinct that is made available for development of housing and employment buildings, including lots, all connector and local streets. It is the total precinct area minus open space, encumbered land and arterial roads. Any additional small local parks defined at the subdivision stage are included in the NDA.

The NDA for the DCP has been calculated in Table 8 to ensure the levies are properly apportioned.

Table 6 Summary Land Use Budget

DESCRIPTION	BROMPTON LODGE PSP		
	HECTARES	% OF TOTAL	% OF NDA
TOTAL PRECINCT AREA (ha)	107.82		
TRANSPORT			
Intersection flaring	3.82	3.54%	4.89%
Ballarto Road Widening	0.23	0.21%	0.30%
Proposed Westernport Highway Widening	9.78	9.07%	12.52%
Sub-total Transport	13.83	12.8%	17.71%
OPEN SPACE			
SERVICED OPEN SPACE			
Waterway and Drainage Reserve	10.79	10.00%	13.81%
Conservation Protected Habitat	0.67	0.62%	0.86%
Tree Reserve	0.25	0.23%	0.32%
Sub-total Service Open Space	11.71	10.86%	14.98%
CREDITED OPEN SPACE			
Local Network Park (Via Cl 52.01)	4.16	3.9%	5.32%
Sub-total Credited Open Space	4.16	3.9%	5.32%
Total All Open Space	15.87	14.7%	20.31%
TOTAL NET DEVELOPABLE AREA (NDA)Ha	78.12	72.46%	
NET DEVELOPABLE AREA - RESIDENTIAL (NDA - R)	78.12	72.46%	

DESCRIPTION	BROMPTON LODGE PSP		
RESIDENTIAL	NDA (HA)	DWELL / NDHA	DWELLINGS
Total Residential Yield Against NDA	78.12	19.0	1,484
Anticipated population @ 2.8 persons per dwelling			4,156

3.1.2 Land Budget and Demand Units

Net developable hectare is the demand unit for this DCP.

Based on the Brompton Lodge PSP, there is one development type included in this DCP: Urban Development. Urban Development is defined broadly to include all forms of development, including residential subdivision and development within the local town centres. Urban Development also includes any non-residential uses within the residential area such as a place of worship, education centre, retirement village, nursing home, child care centre, medical centre or convenience store or any other approved use. There is a total of 78.13 Net Developable Hectares in the DCP area.

It is important to note that the number of Net Developable Hectares (that is the demand units) is based on the land budgets in Table 6.

The 'per Net Developable Hectare' contributions will not and must not be amended to respond to minor changes to land budgets that may result from the subdivision process. In other words, the DCP is permanently linked to the calculation of Net Developable Area set out in the detailed land budget in Table 8.

For the purposes of the DCP, the number of developable hectares will only change if the Collecting Agency agrees to a variation to the Precinct and detailed land budget and associated tables.

The property specific land budget included within each PSP should be used to determine the number of developable hectares (for DCP purposes) on individual parcels.

3.1.3 Property Specific Calculations

The NDA in the summary land use budget (Table 6) used to calculate the contributions within the DCP has been based on an assessment of land use within individual properties. The property specific land budget included within each PSP should be used to determine the number of developable hectares (for DCP purposes) on individual parcels.

The property specific NDA and rates listed in this DCP determine the total contribution for each property.

3.1.4 Public Open Space Contributions

Overall public open space for the precinct NDA is 5.32%.

This DCP contributes towards the purchase of land for local sports fields and construction of sporting surfaces. Local parks are provided through Clause 52.01 of the Scheme and no contribution for local parks is included in this DCP.

Works that a landowner must undertake prior to the handover of land to Council can be considered to be eligible for a partial works in kind credit as per Section 5.3.3 where these works are not considered to be temporary works.

3.2 Calculations of Contributions Charges

3.2.1 Calculation of Costs

Each project has been assigned a land and / or construction cost. These costs are listed in Table 7 and 8. The costs are expressed in 2015 dollars and will be indexed in accordance with the indexation method specified in Section 4.3.

3.2.2 Road Construction and Intersection Works

Cardno have prepared the estimated costs for the transport projects within the DCP. The scope for intersection projects was established collaboratively with Vic Roads in consideration of traffic advice provided by Cardno.

Road construction and intersection costs were derived from concept functional designs which were overlaid onto topographical data and existing conditions in an AutoCAD system. This enables the provision of accurate and detailed component costing as shown on the data sheets in Section 7.0.

Rates for the works have been established by using current road construction estimation rates as of October 2015.

The intersection layout was agreed with the relevant road authority as were the scope of works. The general assumptions used were:

- No land acquisition costs have been allowed for unless stated (these are separately identified in each DCP project costing in Table 7).
- No trunk services have been allowed for.
- Drainage allowance is for 'road reserve or project land' areas i.e. no external catchments.
- A standard excavation depth has been allowed for. Final pavement requirements will be determined at construction stage responding to actual ground conditions.
- Where required an allowance has been made for existing services adjustment or relocation (e.g. electricity poles, water fittings, manholes etc)
- Additional percentage based costs tailored to each individual project have been included for:
    - Traffic management.
    - Field survey and detail design fees.
    - Construction overheads and supervision.
    - Contingency.

The level of contingency for each project reflects the level of design resolution achieved at the time the DCP was prepared. The costs are fully detailed including, for example, street trees, road line marking, footpaths and street lighting.

With respect to road / intersection construction along existing road alignments:

- Most difficulties along the alignment will have been dealt with or built around due to the existing road, therefore upgrading of the pavement will have no effect or only a minor effect on in-ground or underground assets.
- Similar vertical alignments have been allowed for following natural terrain which removes the possibility of extra earthworks being required.
- Higher levels of traffic management and service relocation have been allowed for.
- With respect to road / intersection construction for green field alignments:
- Design generally follows natural terrain.
- Existing service alterations have been included and would be minimal.

The Ballarto Road widening as funded by this DCP has been designed to minimise any redundant works that are required in the ultimate delivery of a future arterial road.

3.2.3 Recreation works

The concept and costing for each facility is based on cost estimates previously prepared by Casey City Council and based on comparisons with other similar facilities.

3.2.4 Community Centre Projects

The concept and costing for each facility is based on cost estimates previously prepared for the Clyde DCP and by Casey City Council for the Cranbourne West proposed facilities.

3.2.5 Interim and Temporary works

Interim works (expected to have a lifetime of at least 4 years) are not allowed as costs against this DCP unless expressly listed in the DCP, and nor will temporary works (expected to have a lifetime of less than four years), unless to the satisfaction of the Responsible Authority.

3.2.6 Valuation of Land

The area of land to be acquired through the DCP was identified through information drawn from the Brompton Lodge PSP. A description of the area of land was provided to Herron Todd White as a registered valuer to prepare a valuation which determined the value for each area of land required by the DCP.

Each parcel where land is required for a DCP project has been individually assessed using a 'before and after' methodology to ensure fair compensation for each affected land owner. These values have been used to calculate the cost of the land component of all projects included in the DCP.

3.2.7 Concept designs and cost sheets

Concept designs and cost sheets have been prepared for each transport project.

This information is included in the Appendices.

3.2.8 Cost Apportionment

The DCP apportions a charge to new development according to its expected use of identified infrastructure items. Since development contribution charges are levied 'up-front', a measure of actual use by individual development sites is not possible. Therefore costs must be shared in accordance with the estimated share of use.

This DCP calculates what development should pay towards provision of the identified infrastructure items. This is the total cost of the item (after deducting other funding sources and making allowance for any external demand) divided by total demand units.

The balance of the cost of the items not recovered under this DCP will be funded from alternative sources.

3.2.9 Non-Government Schools

The development of land for a non government school is exempt from the requirement to pay a development infrastructure levy and a community infrastructure levy under this Development Contribution Plan.

3.2.10 Schedule of Costs

Table 7 calculates the amount of contributions payable by the PSP area for each infrastructure category.

3.2.11 Summary of Charges per Hectare

Table 7 shows the quantum of funds to be contributed = towards each infrastructure project. This adds up to the total amount of funds recoverable under the DCP.

Table 1 provides a snapshot of the key costs and charges.

Table 7 Infrastructure Project Timing and Calculation of Costs - Development Infrastructure Levy

DCP PROJECT NUMBER	PROJECT	INFRASTRUCTURE CATEGORY	LAND AREA (HA)	ESTIMATED PROJECT COSTS			% APPORTIONED TO DCP (INTERNAL USE)	TOTAL COST RECOVERED BY DCP	RESIDENTIAL - CONTRIBUTION PER NDHA-R
				LAND	CONSTRUCTION	TOTAL			
	Road Projects								
RD-01	Ballarto Road widening	Development	0.23	$230,810	$5,117,435	$5,348,245	100%	$5,348,245	$68,458
RD-02	Chevron Avenue upgrade	Development	N/A	$0	$732,056	$732,056	50%	$366,028	$4,685
	Sub-total road projects		0.23	$230,810	$5,849,491	$6,080,301		$5,714,273	$73,143
	Intersection Projects								
IN-01	Ballarto Road and Woodlands Road	Development	0.93	$720,555	$1,488,102	$2,208,657	100%	$2,208,657	$28,271
IN-02	Ballarto Road and Eastern connector	Development	1.01	$963,085	$0	$963,085	100%	$963,085	$12,328
IN-03	Cranbourne-Frankston Road and Woodlands Road	Development	1.04	$714,191	$3,880,046	$4,594,237	100%	$4,594,237	$58,807
IN-04	Cranbourne-Frankston Road and Eastern connector	Development	0.60	$395,809	$2,734,934	$3,130,743	100%	$3,130,743	$40,074
IN-05	Ballarto Road/Cranbourne-Frankston Road/Pearcedale Road	Development	0.24	$255,550	$950,799	$1,206,349	100%	$1,206,349	$15,441
IN-06	Ballarto Road/Westerport Highway	Development	N/A	$0	$628,296	$628,296	100%	$628,296	$8,042
	Sub-total intersection projects		3.82	$3,049,190	$9,682,177	$12,731,367		$12,731,367	$162,963
	Community facilities								
CO-01	Family and Childrens Centre- 1/630 Hall Road	Development		$1,250,000	$4,104,000	$5,354,000	52%	$2,781,403	$35,602
CO-02	Community Centre- Cranbourne West Activity Centre	Development			$7,000,000	$7,000,000	14%	$1,001,431	$12,818
	Sub-total community facilities			$1,250,000	$11,104,000	$12,354,000		$3,782,834	$48,421
	Local Sports Reserves								
LS-01	Local sports reserve at 1070 Cranbourne-Frankston Road, Cranbourne.	Development	6.60	$7,320,000	$2,997,623	$10,317,623	37%	$3,791,069	$48,526
	Sub-total sports reserves (development infrastructure projects)		6.60	$7,320,000	$2,997,623	$10,317,623		$3,791,069	$48,526
	Summary								
	Total cost all projects							$26,019,544	
	Total Development Infrastructure Levy per NDA								**$333,053**

Table 8 Infrastructure Project Timing and Calculation of Costs- Community Infrastructure Levy

DCP PROJECT NUMBER	PROJECT	INFRASTRUCTURE CATEGORY	ESTIMATED PROJECT COST: CONSTRUCTION	TOTAL ESTIMATED PROJECT COST: LAND & CONSTRUCTION	% APPORTIONED TO DCP (INTERNAL USE)	TOTAL COST ATTRIBUTED TO DCP
P-01	Pavillions within sports reserve at 1070 Cranbourne-Frankston Road, Cranbourne	Community	$2,868,068	$2,868,068	36%	$1,024,169
	Sub-total sports reserves (community infrastructure levy)		$2,868,068	$2,868,068		$1,024,169
	Summary					
	Total Community Infrastructure Levy per Dwelling					$690
	Total Community Infrastructure Levy Estimated Raised Via Brompton Lodge DCP					$1,024,169

4.0 DEVELOPMENT CONTRIBUTIONS PLAN ADMINISTRATION

This section sets out how this DCP will be administered and covers the timing of payment, provision of works and land in kind and how funds generated by this DCP will be managed in terms of reporting, indexation and review periods.

The DCP Development Infrastructure Levy applies to subdivision and / or development of land. The DCP Community Infrastructure Levy applies to the construction of dwellings and must be paid prior to the issue of a Building Permit.

Council will be both the Collecting Agency and the Development Agency for the purposes of this DCP.

4.1 Payment of Contribution Levies and Payment Timing

4.1.1 Development Infrastructure

For subdivision of Land

- A development infrastructure levy must be paid to the Collecting Agency for the land within the following specified time, namely after certification of the relevant plan of subdivision but not more than 21 days prior to the issue of a Statement of Compliance with respect to that plan or included in an implementation agreement under Section 173 of the Planning and Environment Act 1987.
- Where the subdivision is to be developed in stages, the infrastructure levy for the stage to be developed may only be paid to the Collecting Agency within 21 days prior to the issue of a Statement of Compliance for that stage provided that a Schedule of Development Contributions is submitted with each stage of plan of subdivision. This Schedule must show the amount of the development contributions payable for each stage and the value of the contributions for prior stages to the satisfaction of the Collecting Agency or included in an implementation agreements under Section 173 of the Planning and Environment Act 1987.

If the Collecting Agency agrees to works or provision of land in lieu of the payment of the infrastructure levy, the land owner must enter into an agreement under Section 173 of the Planning and Environment Act 1987 in respect of the proposed works or provision of land in lieu to specific requirements.

For development of land where no subdivision is proposed

- Provided a development infrastructure levy has not already been paid on subject land, an infrastructure levy must be paid to the Collecting Agency in accordance with the provisions of the approved DCP for each demand unit (Net Developable Hectare) proposed to be developed prior to the commencement of any development (i.e. development includes buildings, car park, access ways landscaping and ancillary components). The Collecting Agency may require that contributions be made at either the planning or building permit stage for Development Infrastructure.

If the Collecting Agency agrees to works or provision of land in lieu of the payment of the infrastructure levy, the land owner must enter into an agreement or other suitable arrangement under Section 173 of the Planning and Environment Act 1987 in relation to the proposed works or land in lieu.

Where no planning permit is required

The following requirements apply where no planning permit is required. The land may only be used and developed subject to the following requirements being met:

- Unless otherwise agreed to by the Collecting Agency in a Section 173 agreement, a development infrastructure levy must be paid to the Collecting Agency prior to the commencement of any development in accordance with the provision of this approved Development Contributions Plan for the land.

If the Collecting Agency agrees to works or provision of land in lieu of the payment of the infrastructure levy, the land owner must enter into an agreement under Section 173 of the Planning and Environment Act 1987 in respect of the proposed works or provision of land in lieu.

4.1.2 Community Infrastructure Levy

Contributions relating to community infrastructure are to be made by the home builder prior to the issue of a building permit. However, development proponents are encouraged to pay the levy prior to the issue of a statement of compliance to reduce the administrative burden of collection from individual home builders.

Levies for 'residential buildings' will be calculated at the rate for a single dwelling. In all other forms of accommodation, the dwelling is the individual unit (such as each dwelling in a residential village, retirement village, serviced apartment complexes and so on). Corrective institutions are exempt.

A community infrastructure levy is not payable for a dwelling on a lot which was created prior to the date that this DCP was first incorporated into the Casey Planning Scheme.

4.1.3 Works in Kind

The Collecting Agency may permit development proponents to undertake works in lieu of cash payments, providing that:

- The works constitute project(s) funded by this DCP.
- The Collecting Agency agrees that the timing of the works would be consistent with priorities in this DCP.
- The development proponent complies with appropriate tendering, documentation, supervision and related provisions.
- The works are defined and agreed in a Section 173 agreement.
- Works must be provided to a standard that accords with this DCP to the satisfaction of the Development Agency, unless an alternative is agreed by both the Development Agency and the Collecting Agency.
- Detailed design must be approved by the Development Agency and the Collecting Agency and must generally accord with the expectations outlined in this DCP unless an alternative is agreed by both the Development Agency and the Collecting Agency.
- The construction of works must be completed to the satisfaction of the Development Agency and the Collecting Agency.
- There should be no negative financial impact on this DCP to the satisfaction of the Collecting Agency.
- In particular, the works will only be accepted in lieu of a financial contribution required by this DCP to the extent that they constitute part or all of the design of the infrastructure item and reduce the cost to complete that design, to the Collecting Agency's satisfaction. Temporary works will not be accepted as works in kind.

Where a Collecting Agency agrees that works are to be provided by a development proponent in lieu of cash contributions (subject to the arrangements specified above):

- The credit for the works (unless an alternative approach is agreed with the Collecting Agency) provided shall equal the final cost of the works up to the maximum identified in this Development Contributions Plan, taking into account the impact of indexation, or to an alternative figure approved by the Collecting Agency.
- The value of the works provided in accordance with the principle outlined above, will be off-set against the development contributions liable to be paid by the development proponent.
- No further financial contributions will be required until the agreed value of any credits is used.

4.1.4 Credit for Over Provision

Where the Collection Agency agrees that a development proponent can physically provide an infrastructure item (either works and/or land) the situation may arise where the developer makes a contribution with a value that exceeds that required by the DCP for the individual project.

In such a case the developer may be entitled to credits against other projects in the DCP to the extent of the excess contribution. Alternatively, the developer may seek an agreement with the Collecting Agency to provide for a cash reimbursement where a significant over contribution has been made on a particular project.

The details of credits and reimbursements will need to be negotiated with, and agreed to by the Collecting Agency.

4.1.5 Land uses exempt from contributions

Where land is subdivided or developed for the purpose of a government school, non government school or any other use that is partly or wholly exempt from development contributions and the land is subsequently used for a purpose other than as one of those exempt uses, the owner of that land must pay to the Collecting Agency development contributions in accordance with the provisions of the DCP. The development infrastructure levy and where applicable, the community infrastructure levy must be paid within 28 days of the date of the commencement of the construction of any buildings or works for that alternative use.

4.2 Funds Administration

The administration of the contributions made under this DCP will be transparent. Details of funds received and expenditures will be held by the Collecting Agency in accordance with the provisions of the Local Government Act 1993 and the Planning and Environment Act 1987.

The administration of contributions made under this DCP will be transparent and demonstrate:

- The amount and timing of funds collected.
- The sources of the funds collected.
- The amount and timing of expenditure on specific projects.
- The project on which the expenditure was made.
- The account classes or individual project classes.
- Details of any works-in-kind arrangements for project provision.
- Any pooling or quarantining of funds to deliver specific projects where applicable.

The Collecting Agency will provide for regular monitoring, reporting and review of the monies received and expended in accordance with this DCP.

The Collecting Agency will establish interest bearing accounts and all monies held in these accounts will be used solely for the provision of infrastructure as specified in this DCP, as required under Section 46QB(2) of the Planning and Environment Act 1987.

Should the Development Agency achieve savings on any project, or resolve not to proceed with any of the infrastructure projects listed in this DCP, the funds collected for these items will be used for alternative works in the same infrastructure class as specified by this DCP. Such funds may also be used for the provision of additional works, services or facilities where approved by the Minister responsible for the Planning and Environment Act, or will be refunded to developers and / or owners of land subject to these infrastructure charges.

4.3 Construction and Land Value Costs Indexation

Capital costs of all infrastructure items (with the exception of land) are in 2015 dollars and will be indexed by the Collecting Agency quarterly to take account of inflation.

In relation to the costs of infrastructure items other than land, the cost must be adjusted according to the following methods:

- Roads, intersections and bridges – in line with the Australian Bureau of Statistics Producer Price Indexes, Road and Bridge Construction Index, Victoria.
- All other infrastructure items - in line with the Australian Bureau of Statistics Producer Price Indexes, Non-Residential Building Construction Index, Victoria.

Land values will be re-valued annually by a registered valuer based on a 'before and after' methodology for each lot that includes land for a DCP project and where relvant a site specific methodology will be used for community facilities and sports reserves outside of the precinct.

Community Infrastructure Levy projects are not indexed as the value of the contribution is set by the Planning and Environment Act 1987.

If in the future the Community Infrastructure Levy is amended, then the levy applicable to the release of any remaining dwellings may be adjusted in accordance with the revised legislative and regulatory approach as directed by the Minister for Planning.

4.4 Development Contributions Plan Review Period

This DCP adopts a long-term outlook for development. It takes into account planned future development in the Brompton Lodge area. A 'full development' horizon of land within the current Urban Growth Boundary within 30 years of gazettal of this DCP has been adopted.

This DCP commences on the date of incorporation into the Casey Planning Scheme. This DCP will end when development within the DCP area is complete, which is projected to be 30 years after gazettal, or when the DCP is removed from the Planning Scheme.

The DCP is expected to be revised and updated every 5 years (or more frequently if required). This will require an amendment to the Casey Planning Scheme to replace this document with an alternative, revised document. Any review will need to have regard to any arrangements (for example an agreement under s173 of the Act) for the implementation of this DCP.

4.5 Collecting Agency (Agency Responsible for Collecting Infrastructure Levy)

Casey City Council is the Collecting Agency pursuant to section 46K(1)(fa) of the Planning and Environment Act 1987 which means that it is the public authority to whom all levies are payable. As the Collecting Agency, Casey City Council is also responsible for the administration of this DCP and also its enforcement pursuant to Section 46QC of the Act.

4.6 Development Agency (Agency Responsible for Works)

Casey City Council is the Development Agency and is responsible for the provision of all of the DCP projects identified in this DCP.

5.0 IMPLEMENTATION STRATEGY

This section provides further details regarding how the Collecting Agency intends to implement this DCP. In particular, this section clearly identifies the rationale for the implementation strategy and the details the various measures that have been adopted to reduce the risk posed by this DCP to all parties.

4.7 Rationale for the Implementation Strategy

This implementation strategy has been incorporated into this DCP to provide certainty to the Collecting Agency and development proponents. The implementation strategy recognises the complexities associated with infrastructure provision and funding and seeks to minimise risk to the Collection Agency, Development Agency, development proponent and future community.

The implementation strategy has been formulated by:

- The PSP (in preparation or in draft).
- Having regard to the development context.
- Assessing the need for finance requirements – up front financing and pooling of funds.
- Agreeing the land value and indexing it appropriately (where possible).
- Identifying preferred implementation mechanisms to achieve the above outcomes and reducing the risk associated with the DCP to ensure that it will be delivered as intended.
- Provision of adequate resources to administer this DCP.

4.8 Implementation Mechanism

Under Section 46P of the Act, the Collecting Agency may accept (with the consent of the Development Agency where the Collecting Agency is not also the Development Agency), the provision of land, works, services or facilities by the applicant in part of full satisfaction of the amount of levy payable. This can be agreed with the Collecting Agency before or after the application for the permit is made or before the development is carried out.

To coordinate the provision of infrastructure, it is proposed that the owner of all property enter into an agreement under s173 of the Act to provide for provision of works in kind.

Through the approval of these agreements Casey City Council (acting as the Collecting Agency) will consider if and what infrastructure should be provided as in-kind works under this DCP in accordance with Section 46P of the Act. The agreement must include a list of the DCP infrastructure which the Collecting Agency has agreed in writing to allow to be provided as works in kind.

Table 9 External Apportionment

DCP PROJECT NUMBER	PROJECT	% APPORTIONED TO DCP (INTERNAL USE)	CRANBOURNE WEST DCP	SURROUNDING DEVELOPMENT (NRCL, RANFURLIE GOLF, AMSTEL)	CASEY CITY COUNCIL
CO-01	Family and Childrens Centre- 1/630 Hall Road	52%	48%		
CO-02	Community Centre- Cranbourne West Activity Centre	14%	31%	55%	
LS-01	Local sports reserve at 1070 Cranbourne-Frankston Road, Cranbourne.	37%		63%	
P-01	Pavillions within sports reserve at 1070 Cranbourne-Frankston Road, Cranbourne	36%		64%	
RD-02	Chevron Avenue Upgrade	50%			50%

5.0 OTHER INFORMATION

5.1 Acronyms

'the Act'	Planning and Environment Act 1987
AHD	Australian Height Datum
AFL	Australian Football League ovals
CAD	Central Activities District
CIL	Community Infrastructure Levy
DCP	Development Contributions Plan
DEECD	Department of Education & Early Childhood Development
DIL	Development Infrastructure Levy
DoT	Department of Transport
ENDA	Employment Net Developable Area
GAA	Metropolitan Planning Authority
GDA	Gross Developable Area
Ha	Hectare
LTC	Local Town Centre
MCA	Main Catchment Area
MCH	Maternal & Child Health
MSS	Municipal Strategic Statement
MTC	Major Town Centre
NDA	Net Developable Area
NDHa	Net Developable Hectare
PPTN	Principle Public Transport Network
PSP	Precinct Structure Plan
PTC	Principle Town Centre
P-6	School Prep to Year 6
P-12	State School Prep to Year 12
RNDA	Residential Net Developable Area
Sqm	Square Metres
UGB	Urban Growth Boundary
UGZ	Urban Growth Zone

5.2 Glossary

Active Open Space

Land set aside for the specific purpose of formal/organised club based sports.

Activity Centre

See 'Town Centre'.

Arterial Road

A higher order road providing for moderate to high volumes at relatively high speeds typically used for inter-suburban journeys and linking to freeways, and identified under the Road Management Act 2004. All declared arterials are managed by the State Government.

Co-Location

Adjoining land uses to enable complementary programs, activities and services and shared use of resources and facilities. For example, the co-location of schools and active open space.

Community Facilities

Infrastructure provided by government or non-government organisations for accommodating a range of community support services, programs and activities. This includes facilities for education and learning (e.g. government and non-government schools, universities, adult learning centres); early years (e.g. preschool, maternal and child health, childcare); health and community services (eg. hospitals, aged care, doctors, dentists, family and youth services, specialist health services); community (e.g. civic centres, libraries, neighbourhood houses); arts and culture (e.g. galleries, museums, performance space); sport, recreation and leisure (e.g. swimming pools); justice (e.g. law courts); voluntary and faith (e.g. places of worship) and emergency services (e.g. police, fire and ambulance stations).

Connector Street

A lower order street providing for low to moderate volumes and moderate speeds linking local streets to the arterial network Managed by the relevant local council. (See Table C1 in clause 56)

Development Contributions Plan

Document that sets out the contributions expected from each individual landowner to fund infrastructure and services. Refer to Part 3B of the Planning and Environment Act l987.

Encumbered Land

Land that is constrained for development purposes. Includes easements for power/ transmission lines, sewers, gas, waterways, drainage, retarding basins/wetlands, landfill, conservation and heritage areas. This land may be used for a range of activities (e.g. walking trails, sports fields).

Freeway

A high speed and high volume road with the highest level of access control and typically used for longer distance journeys across the metropolitan area and country Victoria. All freeways are managed by VicRoads.

Growth Area

Areas on the fringe of metropolitan Melbourne around major regional transport corridors that are designated for large-scale change, over many years from rural to urban use. Melbourne has six growth areas called Casey-Cardinia; Hume; Melton-Caroline Springs; Whittlesea, Wyndham and Mitchell.

Growth Corridor Plan

Government document that sets long-term strategic planning direction to guide the creation of a more sustainable community in the growth areas.

High Density Housing

Housing with an average density of more than 30 dwellings per net developable hectare.

Housing Density (Net)

The number of houses divided by net developable area

Linear Open Space Network

Corridors of open space, mainly along waterways that link together, forming a network.

Land Budget Table

A table setting out the total precinct area, net developable area and constituent land uses proposed within the precinct.

Lot

A part (consisting of one or more pieces) of any land (except a road, a reserve, or common property) shown on a plan, which can be disposed of separately and includes a unit or accessory unit on a registered plan of strata subdivision and a lot or accessory lot on a registered cluster plan.

Lower Density Housing

Housing with an average density of less than 10 dwellings per hectare.

Major Employment Area

Areas identified on the Growth Corridor Plan for economic and employment growth.

Major Town Centre

Town centres that have similar characteristics to Principal Town Centres, but serve smaller catchment areas.

Medium Density Housing

Housing with an average density of 16 to 30 dwellings per net developable hectare.

Native Vegetation

Plants that are indigenous to Victoria, including trees, shrubs, herbs, and grasses.

Net Developable Area

Total amount of land within the precinct that is made available for development of housing and employment buildings, including lots, local and connector streets. Total precinct area minus community facilities, schools and educational facilities and open space, arterial roads and encumbered land. Small local parks defined at subdivision stage are included in net developable area.

Passive Open Space

Open space that is set aside for parks, gardens, linear corridors, conservation bushlands, nature reserves, public squares and community gardens that are made available for passive recreation, play and unstructured physical activity including walking, cycling, hiking, revitalisation, contemplation and enjoying nature.

Precinct Structure Plan

A statutory document that describes how a precinct or series of sites within a growth area will be developed over time. A precinct structure plan sets out the broad environmental, social and economic parameters for the use and development of land within the precinct.

Principal Public Transport Network

A high-quality public transport network that connects Principal and Major Activity Cent res, and comprises the existing radial fixed-rail network, extensions to this radial network and new cross-town bus routes.

Public Open Space

Land that is set aside in the precinct structure plan for public recreation or public resort, or as parklands, or for similar purposes. Incorporates active and passive open space.

Town Centre

Provide the focus for services, commercial and retail based employment and social interaction. They are where people shop, work, meet, relax and live. They are well-served by public transport, they range in size and intensity of use. In the growth areas, these are referred to as principal activity centres, major activity centres, neighbourhood activity centres and local centres.

Urban Growth Boundary

A statutory planning management tool used to set clear limits to metropolitan Melbourne's urban development.

Urban Growth Zone

Statutory zone that applies to land that has been identified for future urban development. The UGZ has four purposes: (1) to manage transition of non-urban land into urban land; (2) to encourage development of well-planned and well-serviced new urban communities in accordance with an overall plan; (3) to reduce the number of development approvals needed in areas where an agreed plan is in place; and (4) to safeguard non-urban land from use and development that could prejudice its future urban development.

6.0 APPENDICES

APPENDIX A Precinct Infrastructure Plan

GROUP	PROJECT CATEGORY	PROJECT	WORKS DESCRIPTION	LEAD AGENCY	INCLUDED IN DCP
TRANSPORT					
Transport	Road	Ballarto Road construction	Land and construction - widening where required and construction of the first stage carriageways	City of Casey	YES
Transport	Road	Western Port Highway widening	Upgrade to freeway - Land aquired by PAO	VicRoads	NO
Transport	Road	Upgrade of Chevron Avenue	Upgrade of existing carriageway to an urban standard	City of Casey	YES
Transport	Intersection	Cranbourne-Frankston Road, Chevron Avenue and the North South Connector	Land for ultimate configuration and construction of interim signalised intersection	City of Casey	YES
Transport	Intersection	Cranbourne-FrankstonRoad,Woodlands Road and the North South Connector	Land for ultimate configuration and construction of signalised intersection	City of Casey	YES
Transport	Intersection	Ballarto Road and Woodlands Road	Purchase of land for intersection (ultimate treatment) and construction of roundabout (interim) and pedestrian signals	City of Casey	YES
Transport	Intersection	Ballarto Road and Eastern Connector	Purchase of land for intersection (ultimate treatment)	City of Casey	YES
Transport	Intersection	Ballarto Road and Western Port Highway	Construction of 4th leg of Highway to connector roundabout	City of Casey	YES
Transport	Intersection	Ballarto Road/Cranbourne-Frankston Road/Pearcedale Road	Construction of 5th leg of arterial to connector intersection (interim treatment)	City of Casey	YES
COMMUNITY FACILITIES					
Community	Community	Family and Children's Centre	Contribution towards land and construction of Family and Children's Centre at 1/630 Hall Road, Cranbourne West	City of Casey	YES
Community	Community	Community Centre	Contribution towards construction of the future integrated community centre in the Cranbourne West Activity Centre	City of Casey	YES
Community	Sports Reserve	Recreation facility	Contribution towards land and construction of a recreation facilitiy at 1070 Cranbourne-Frankston, Cranburne	City of Casey	YES

APPENDIX B Property Specific Land Use Budget

		TRANSPORT							SERVICED OPEN SPACE			CREDITED OPEN SPACE		
PSP PROPERTY ID	TOTAL AREA (HECTARES)	BALLARTO ROAD WIDENING	IN-01	IN-02	IN-03	IN-04	IN-05	PROPOSED WESTERNPORT HIGHWAY WIDENING	WATERWAY/DRAINAGE RESERVE	CONSERVATION (PROTECTED HABITAT)	TREE RESERVE	LOCAL PARKS	TOTAL NET DEVELOPABLE AREA (HECTARES)	NET DEVELOPABLE AREA % OF PROPERTY
1	15.26							3.23	3.49			0.00	8.55	56%
2	18.83				0.10			1.31	2.89	0.67	0.14	1.17	12.55	67%
3	40.17				0.95	0.60		5.24	3.86		0.11	1.33	28.07	70%
4	12.07	0.03	0.93						0.05			0.33	10.75	89%
5	8.34			0.87					0.00			0.53	6.93	83%
6	2.54								0.50			0.00	2.04	80%
7	6.77	0.06		0.14					0.00			0.80	5.77	85%
8	0.20								0.00			0.00	0.20	100%
9	3.65	0.15					0.24		0.00			0.00	3.26	89%
SUB-TOTAL	107.82	0.23	0.93	1.01	1.04	0.60	0.24	9.78	10.79	0.67	0.25	4.16	78.12	72%
TOTAL	108	0.23	0.93	1.01	1.04	0.60	0.24	9.78	10.79	0.67	0.25	4.16	78.12	72%

APPENDIX C Interim Intersection Designs

BROMPTON LODGE PSP
INTERIM INTERSECTION DESIGN



DRAWING LIST

V160589-TR-DG-1104	COVER SHEET
V160589-TR-DG-2541	BALLARTO ROAD / WESTERN PORT HIGHWAY
V160589-TR-DG-2542	BALLARTO ROAD / WOODLANDS ROAD
V160589-TR-DG-2543	BALLARTO ROAD / EASTERN CONNECTOR ROAD
V160589-TR-DG-2544	BALLARTO ROAD / CRANBOURNE - FRANKSTON ROAD / PEARCEDALE ROAD
V160589-TR-DG-2545	CRANBOURNE - FRANKSTON ROAD / WOODLANDS ROAD
V160589-TR-DG-2546	CRANBOURNE - FRANKSTON ROAD / CHEVRON ROAD / EASTERN CONNECTOR ROAD

N

MELWAY MAP REF 132 & 133

Rev	Date	Description	Drawn	Appr.
2	25.07.16	MPA COMMENTS	EG	BM
1	20.07.16	ISSUED FOR INFORMATION	EG	BM

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0 25 50 100
1:5000 @ A1

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2. CADASTRAL INFORMATION OBTAINED FROM VICMAP DATA.
3. DECLARED STATE HIGHWAY - WESTERN PORT HIGHWAY - SPEED ZONE 80KM/HR
DECLARED MAIN ROAD - FRANKSTON-CRANBOURNE RD - SPEED ZONE 80KM/HR
LOCAL ROAD - BALLARTO ROAD - SPEED ZONE 60KM/HR
4. LIDAR BASE OBTAINED FROM MPA.
5. CRANBOURNE - FRANKSTON ROAD AND SURROUNDS BASE OBTAINED FROM VICROADS.

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Designed	S. McKENZIE
Checked	K. SHAO
Authorised	B. MENTHA

Client	METROPOLITAN PLANNING AUTHORITY
Project	BROMPTON LODGE PSP CRANBOURNE SOUTH CITY OF CASEY
Title	COVER SHEET - LOCALITY PLAN INTERIM INTERSECTION DESIGN CONCEPT LAYOUT
Status	PRELIMINARY NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date	20.07.16
Scale	1:5000
Size	A1
Drawing Number	V160589-TR-DG-1104
Revision	2

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BY: Usercell.goodhardt
DATE PLOTTED: 21/07/2016 10:58:53 AM
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Rev	Date	Description	Drawn	Appr.
1	20.07.16	ISSUED FOR INFORMATION	EG	BM

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3. DECLARED STATE HIGHWAY - WESTERN PORT HWY - SPEED ZONE 80KM/HR
   LOCAL ROAD - BALLARTO ROAD - SPEED ZONE 60KM/HR
4. LIDAR BASE OBTAINED FROM MPA.
5. CRANBOURNE - FRANKSTON ROAD AND SURROUNDS BASE OBTAINED FROM VICROADS.

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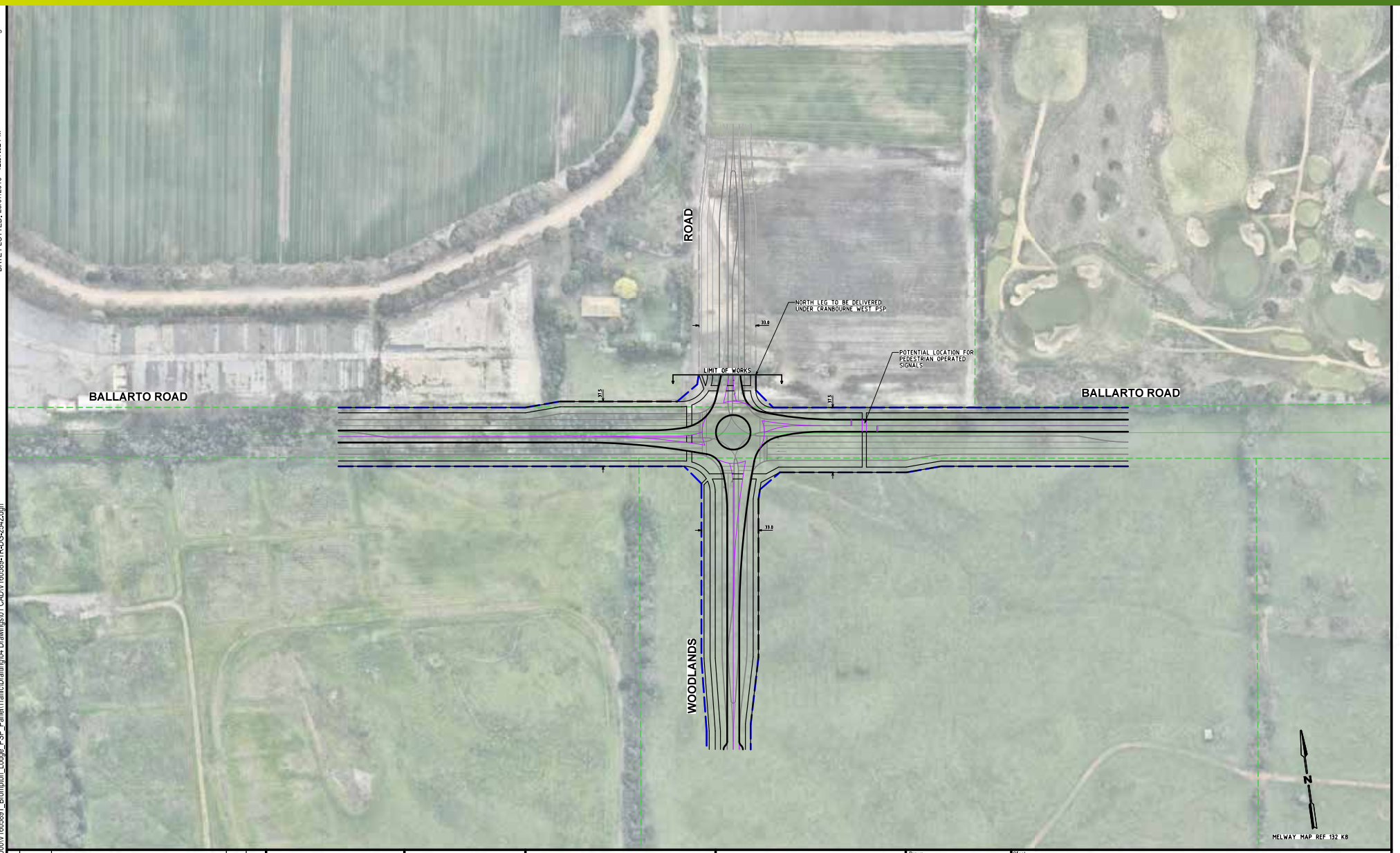
Drawn	E. GOODHARDT
Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: INTERIM INTERSECTION DESIGN
WESTERN PORT HIGHWAY / BALLARTO RD
CONCEPT LAYOUT

Status: **PRELIMINARY**
NOT TO BE USED FOR CONSTRUCTION PURPOSES

Date	Scale	Size
20.07.16	1:1000	A1

Drawing Number: V160589-TR-DG-2541
Revision: 1



BY: Usercell.goodhart
DATE PLOTTED: 25/07/2016 12:57:02 PM
CAD File:M:\2016\0501-1000\V160589T_Brompton_Lodge_PSP_Panell\Traffic\Drafting\04 Drawings\01 CAD\V160589-TR-DG-2542.dgn

Rev.	Date	Description	Drawn	Appr.
2	25.07.16	MPA COMMENTS	EG	BM
1	20.07.16	ISSUED FOR INFORMATION	EG	BM

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GENERAL NOTES
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2. CADASTRAL INFORMATION OBTAINED FROM VICMAP DATA.
3. LOCAL ROAD - BALLARTO ROAD - SPEED ZONE 60KM/HR
4. LIDAR BASE OBTAINED FROM MPA.
5. CRANBOURNE - FRANKSTON ROAD AND SURROUNDS BASE OBTAINED FROM VICROADS.

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Checked	B. HARWOOD
Authorised	B. MENTHA

Client	METROPOLITAN PLANNING AUTHORITY
Project	BROMPTON LODGE PSP CRANBOURNE SOUTH CITY OF CASEY
Title	INTERIM INTERSECTION DESIGN BALLARTO RD / WOODLANDS RD CONCEPT LAYOUT
Status	PRELIMINARY NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date	20.07.16
Scale	1:1000
Size	A1
Drawing Number	V160589-TR-DG-2542
Revision	2



BY: User:ell.goodhardt
DATE PLOTTED: 21/07/2016 1:01:55 AM
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Rev	Date	Description	Drawn	Appr.
1	20.07.16	ISSUED FOR INFORMATION	EG	BM

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1:1000 @ A1

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2. CADASTRAL INFORMATION OBTAINED FROM VICMAP DATA.
3. DECLARED MAIN ROAD - FRANKSTON-CRANBOURNE RD - SPEED ZONE 80KM/HR
   LOCAL ROAD - BALLARTO ROAD - SPEED ZONE 60KM/HR
4. LIDAR BASE OBTAINED FROM MPA.
5. CRANBOURNE - FRANKSTON ROAD AND SURROUNDS BASE OBTAINED FROM VICROADS.

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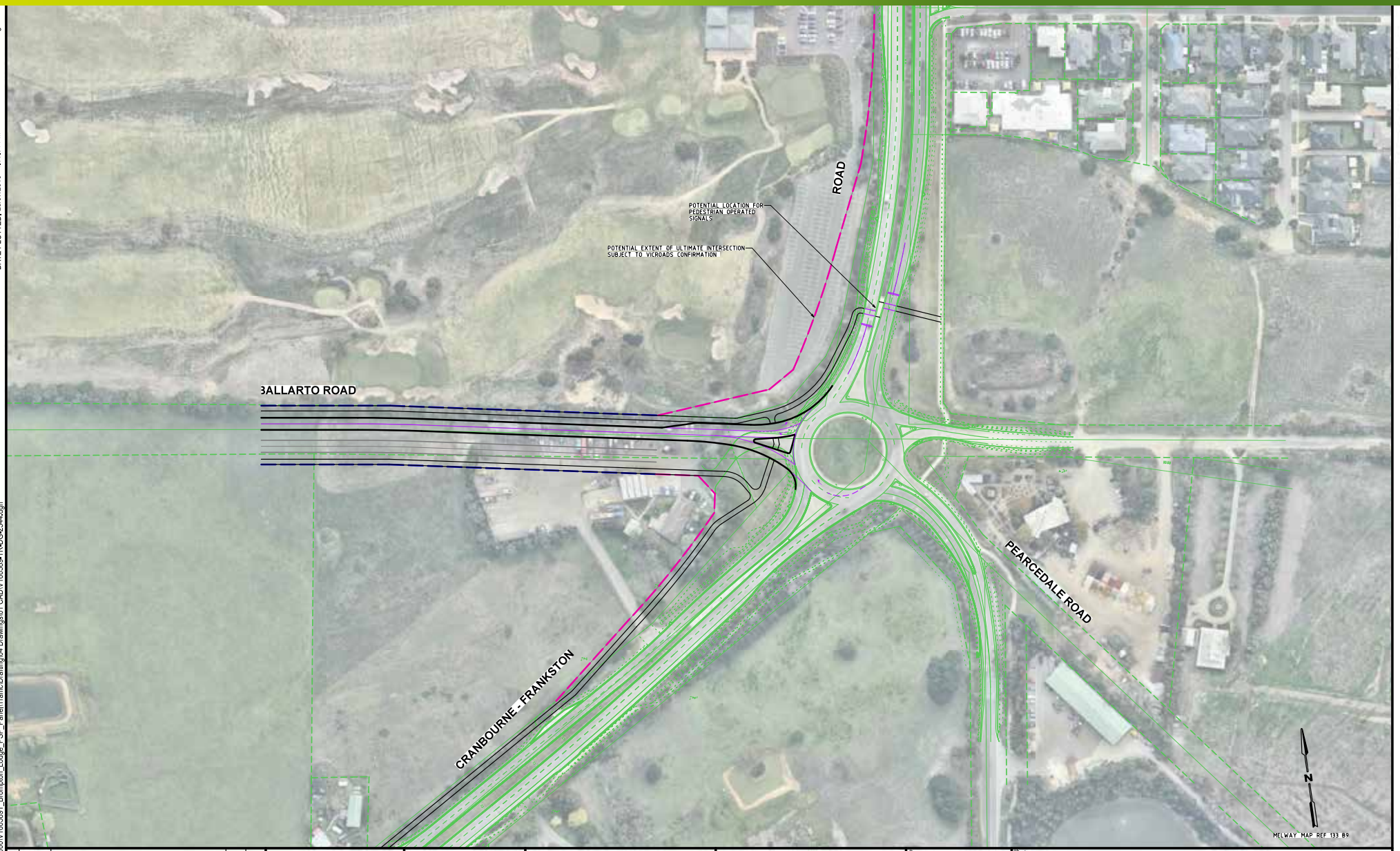
Drawn	E. GOODHARDT
Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: INTERIM INTERSECTION DESIGN
BALLARTO RD / EASTERN CONNECTOR RD
CONCEPT LAYOUT

Status: **PRELIMINARY**
NOT TO BE USED FOR CONSTRUCTION PURPOSES

Date	Scale	Size
20.07.16	1:1000	A1

Drawing Number: V160589-TR-DG-2543
Revision: 1



BY: User:elgoodhar

DATE PLOTTED: 25/07/2016 1:00:04 PM

CAD File:M:\2016\0501-10001\V160589T_Brompton_Lodge_PSP_Panel\Traffic\Drafting\04 Drawings\01 CAD\V160589-TR-DG-2544.dgn

Rev	Date	Description	Drawn	Appr.
2	25.07.16	MPA COMMENTS	EG	BM
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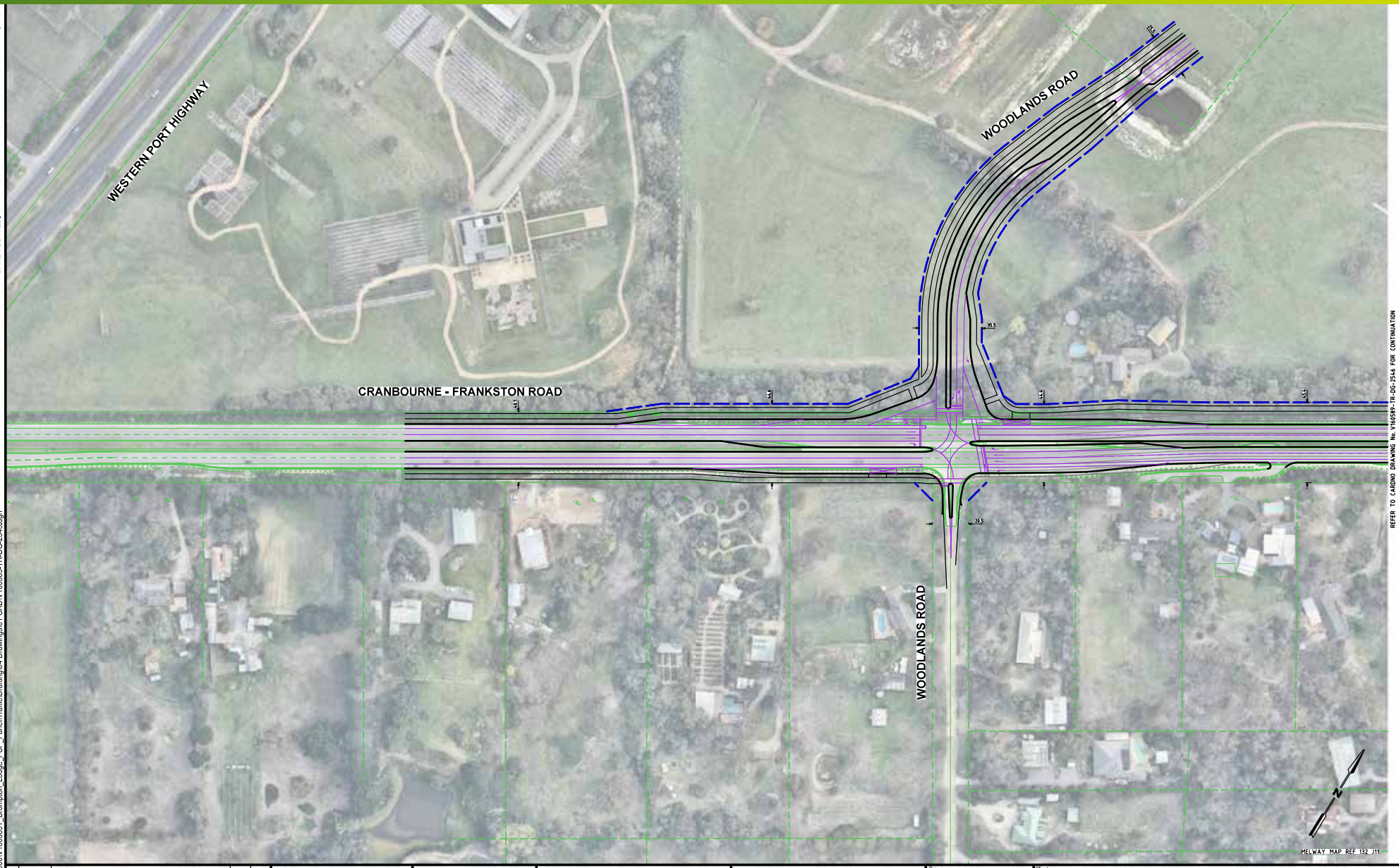
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LOCAL ROAD - BALLARTO ROAD - SPEED ZONE 60KM/HR
4. LIDAR BASE OBTAINED FROM MPA.
5. CRANBOURNE - FRANKSTON ROAD AND SURROUNDS BASE OBTAINED FROM VICROADS.

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Authorised	B. MENTHA

Client	METROPOLITAN PLANNING AUTHORITY
Project	BROMPTON LODGE PSP CRANBOURNE SOUTH CITY OF CASEY
Title	INTERIM INTERSECTION DESIGN BALLARTO RD / CRANBOURNE - FRANKSTON RD CONCEPT LAYOUT
Status	PRELIMINARY NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date	20.07.16
Scale	1:1000
Size	A1
Drawing Number	V160589-TR-DG-2544
Revision	2



BY: User:ell.goodhar
DATE PLOTTED: 21/07/2016 11:04:13 AM
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Rev	Date	Description	Drawn	Appr.
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4. LIDAR BASE OBTAINED FROM MPA.
5. CRANBOURNE - FRANKSTON ROAD AND SURROUNDS BASE OBTAINED FROM VICROADS.

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Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: INTERIM INTERSECTION DESIGN
CARNBOURNE - FRANKSTON RD / WOODLANDS RD
CONCEPT LAYOUT
Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date: 20.07.16
Scale: 1:1000
Size: A1
Drawing Number: V160589-TR-DG-2545
Revision: 1



CAD File:M:\2016\0501-1000\V160589T_Brompton_Lodge_PSP_Panel\Traffic\Drafting\04 Drawings\01 CAD\V160589-TR-DG-2546.dgn
DATE PLOTTED: 21/07/2016 11:04:38 AM
BY: Userelli.goodhar

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1	20.07.16	ISSUED FOR INFORMATION	EG	BM

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1:1000 @ A1

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Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: INTERIM INTERSECTION DESIGN
CRANBOURNE - FRANKSTON RD / EASTERN CONNECTOR RD
CONCEPT LAYOUT

Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES

Date	Scale	Size
20.07.16	1:1000	A1

Drawing Number: V160589-TR-DG-2546
Revision: 1

APPENDIX D Interim Ballarto Road alignment design

BROMPTON LODGE PSP
INTERIM ALIGNMENT DESIGN



DRAWING LIST

V160589-TR-DG-1102	COVER SHEET
V160589-TR-DG-2511	BALLARTO ROAD / WESTERN PORT HIGHWAY
V160589-TR-DG-2512	BALLARTO ROAD / WOODLANDS ROAD AND BALLARTO ROAD / EASTERN CONNECTOR ROAD
V160589-TR-DG-2513	BALLARTO ROAD / CRANBOURNE - FRANKSTON ROAD / PEARCEDALE ROAD

N

MELWAY MAP REF 132 & 133

Rev	Date	Description	Drawn	Appr.
4	25.07.16	MPA COMMENTS	EG	BM
3	20.07.16	MPA COMMENTS	EG	BM
2	15.07.16	ULTIMATE DESIGN OVERLAID FOR COMPARISON	SGM	BM
1	24.06.16	ISSUED FOR INFORMATION	SGM	BM

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Checked	K. SHAO
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: COVER SHEET - LOCALITY PLAN
INTERIM ALIGNMENT DESIGN
CONCEPT LAYOUT

Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES

Date	Scale	Size
24.06.16	1:5000	A1

Drawing Number: V160589-TR-DG-1102
Revision: 4

CAD File: M:\2016\0501-1000\V160589T_Brompton_Lodge_PSP_Panel\Traffic\Drafting\04 Drawings\01 CAD\V160589-TR-DG-1102.dgn
DATE PLOTTED: 25/07/2016 12:55:50 PM
BY: Usersall.goodhardt



BY: Usercell.goodhardt
DATE PLOTTED: 21/07/2016 11:37:03 AM
CAD File: M:\2016\60501-1000\V160589T_Brompton_Lodge_PSP_Panel\Traffic\Drafting\04 Drawings\01 CAD\V160589-TR-DG-2511.dgn

Rev	Date	Description	Drawn	Appr.
3	20.07.16	MPA COMMENTS	EG	BM
2	15.07.16	ULTIMATE DESIGN OVERLAID FOR COMPARISON	SGM	BM
1	24.06.16	ISSUED FOR INFORMATION	SGM	BM

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Designed	S. McKENZIE
Checked	K. SHAO
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: INTERIM ALIGNMENT DESIGN
WESTERN PORT HIGHWAY / BALLARTO ROAD
CONCEPT LAYOUT

Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date: 24.06.16
Scale: 1:1000
Size: A1
Drawing Number: V160589-TR-DG-2511
Revision: 3



Rev	Date	Description	Drawn	Appr.
4	25.07.16	MPA COMMENTS	EG	BM
3	20.07.16	MPA COMMENTS	EG	BM
2	15.07.16	ULTIMATE DESIGN OVERLAID FOR COMPARISON	SGM	BM
1	24.06.16	ISSUED FOR INFORMATION	SGM	BM

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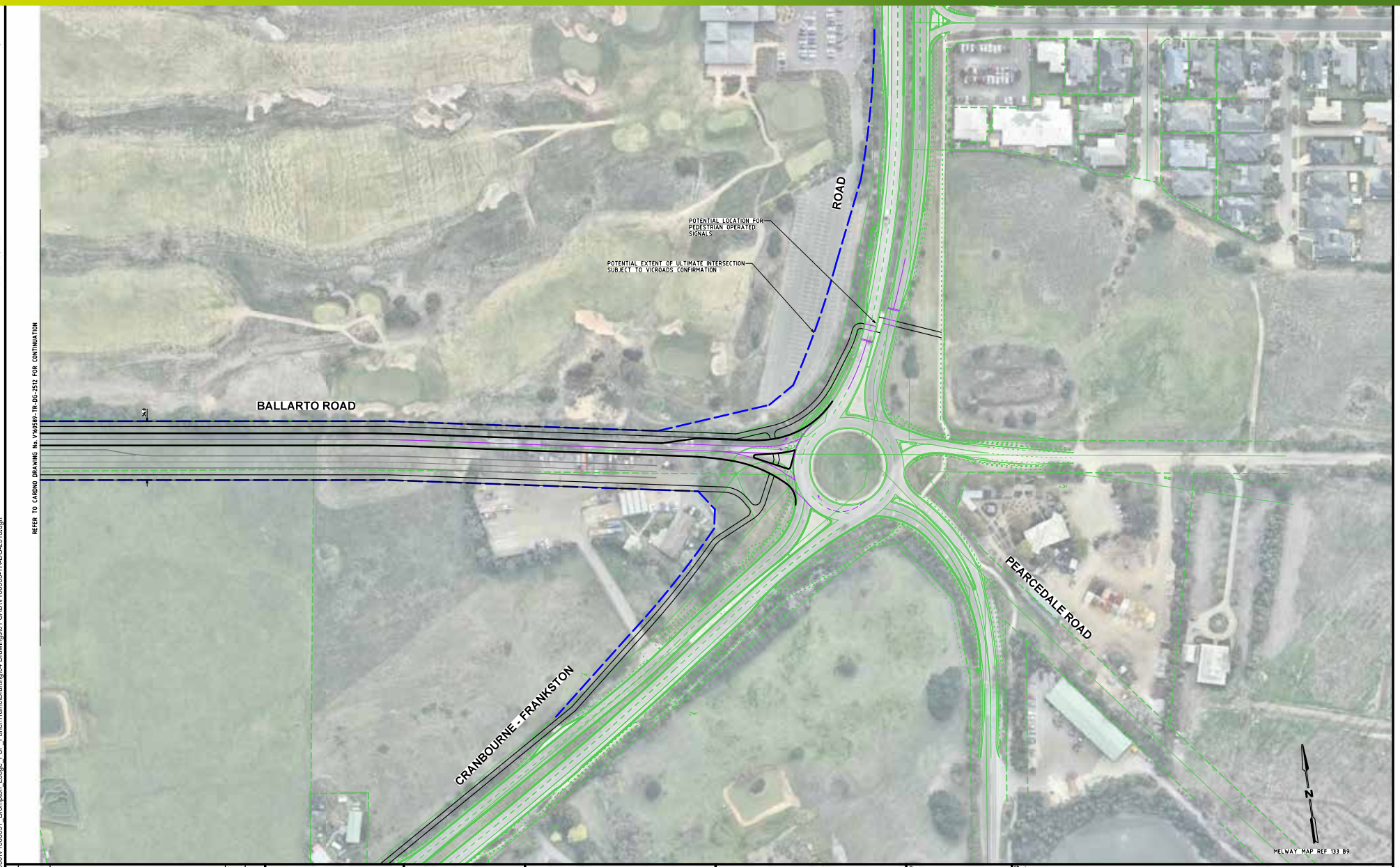
GENERAL NOTES
1. ALL DIMENSIONS TO FACE OF KERB AND CHANNEL U.N.O.
2. CADASTRAL INFORMATION OBTAINED FROM VICMAP DATA.
3. LOCAL ROAD - BALLARTO ROAD - SPEED ZONE 60KM/HR
4. LIDAR BASE OBTAINED FROM MPA.
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Drawn	S. McKENZIE
Designed	S. McKENZIE
Checked	K. SHAO
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date: 24.06.16 | Scale: 1:1000 | Size: A1
Title: INTERIM ALIGNMENT DESIGN
BALLARTO RD / WOODLANDS / EASTERN CONNECTOR
CONCEPT LAYOUT
Drawing Number: V160589-TR-DG-2512 | Revision: 4

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BY: Usersaligoodhardt

DATE PLOTTED: 25/07/2016 12:58:39 PM

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Rev	Date	Description	Drawn	Appr.
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3	20.07.16	MPA COMMENTS	EG	BM
2	15.07.16	ULTIMATE DESIGN OVERLAID FOR COMPARISON	SGM	BM
1	24.06.16	ISSUED FOR INFORMATION	SGM	BM

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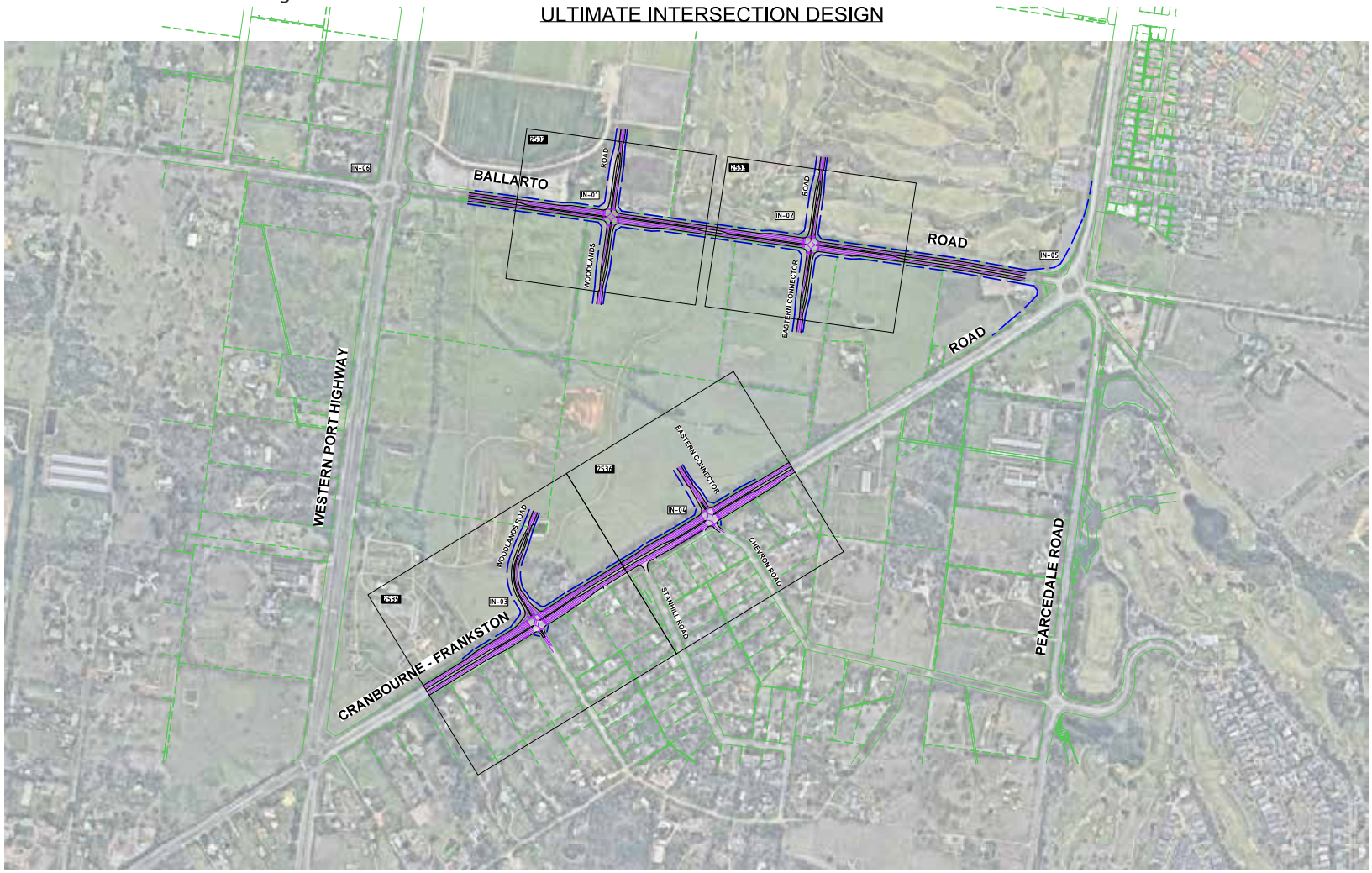
GENERAL NOTES
1. ALL DIMENSIONS TO FACE OF KERB AND CHANNEL U.N.O.
2. CADASTRAL INFORMATION OBTAINED FROM VICMAP DATA.
3. DECLARED MAIN ROAD - FRANKSTON-CRANBOURNE RD - SPEED ZONE 80KM/HR
   LOCAL ROAD - BALLARTO ROAD - SPEED ZONE 60KM/HR
4. LIDAR BASE OBTAINED FROM MPA.
5. CRANBOURNE - FRANKSTON ROAD AND SURROUNDS BASE OBTAINED FROM VICROADS.

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Drawn: S. McKENZIE
Designed: S. McKENZIE
Checked: K. SHAO
Authorised: B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: INTERIM ALIGNMENT DESIGN
BALLARTO ROAD / CRANBOURNE - FRANKSTON ROAD
CONCEPT LAYOUT
Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date: 24.06.16
Scale: 1:1000
Size: A1
Drawing Number: V160589-TR-DG-2513
Revision: 4

BROMPTON LODGE PSP
ULTIMATE INTERSECTION DESIGN



DRAWING LIST

V160589-TR-DG-1103	COVER SHEET
V160589-TR-DG-2532	BALLARTO ROAD / WOODLANDS ROAD
V160589-TR-DG-2533	BALLARTO ROAD / EASTERN CONNECTOR ROAD
V160589-TR-DG-2535	CRANBOURNE - FRANKSTON ROAD / WOODLANDS ROAD
V160589-TR-DG-2536	CRANBOURNE - FRANKSTON ROAD / CHEVRON ROAD / EASTERN CONNECTOR ROAD

N

MELWAY MAP REF 132 & 133

Rev	Date	Description	Drawn	Appr.
1	20.07.16	ISSUED FOR INFORMATION	EG	BM

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1:5000 @ A1

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2. CADASTRAL INFORMATION OBTAINED FROM VICMAP DATA.
3. DECLARED STATE HIGHWAY - WESTERN PORT HIGHWAY - SPEED ZONE 80KM/HR
DECLARED MAIN ROAD - FRANKSTON-CRANBOURNE RD - SPEED ZONE 80KM/HR
- BALLARTO ROAD - SPEED ZONE 80KM/HR
4. LIDAR BASE OBTAINED FROM MPA.
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Drawn	E. GOODHARDT
Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

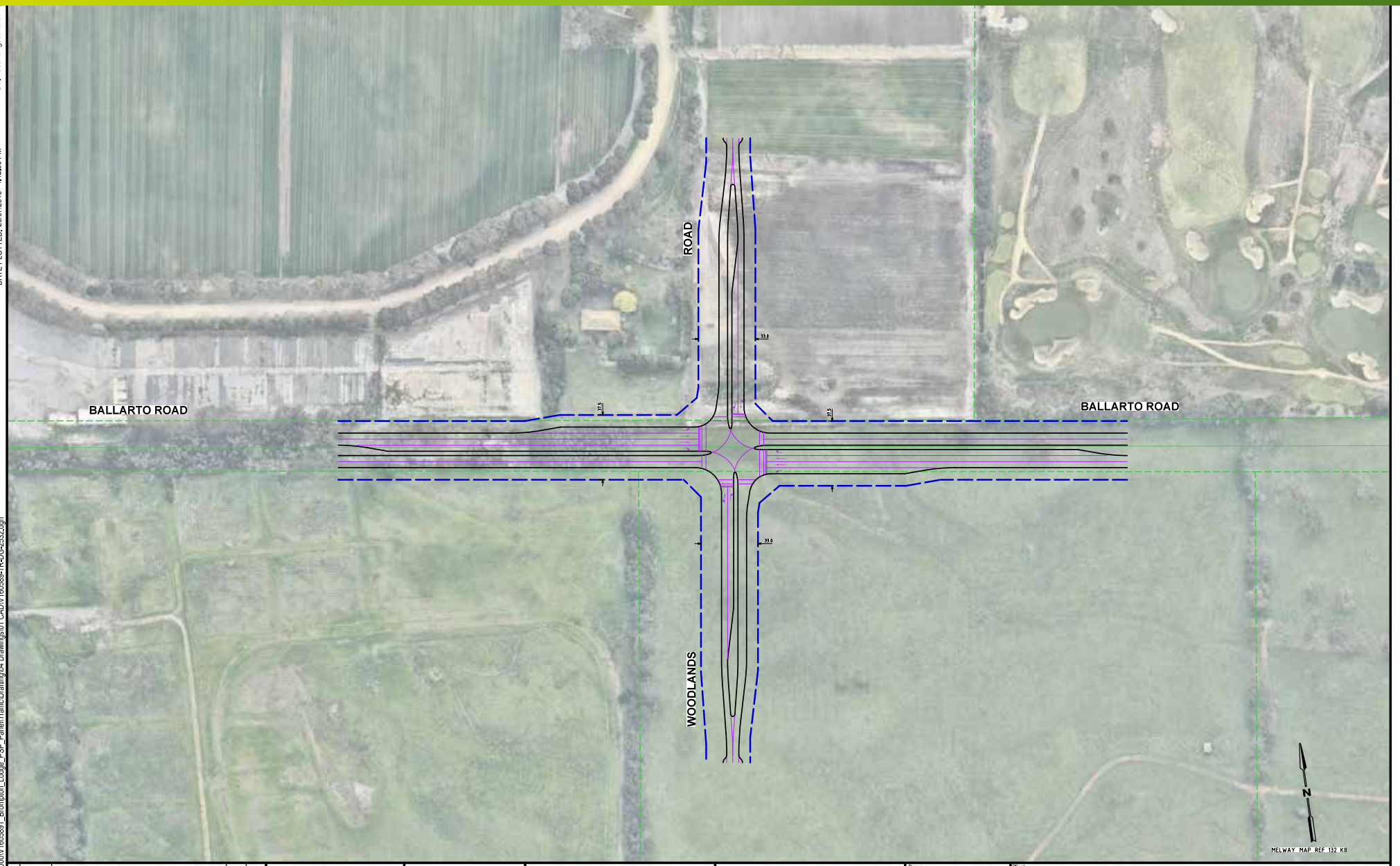
Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: COVER SHEET - LOCALITY PLAN
ULTIMATE INTERSECTION DESIGN
CONCEPT LAYOUT

Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES

Date	Scale	Size
20.07.16	1:5000	A1

Drawing Number: V160589-TR-DG-1103
Revision: 1

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DATE PLOTTED: 21/07/2016 11:10:47 AM
BY: User:ellgoodhardt



BY: Userall.goodhardt
DATE PLOTTED: 20/07/2016 4:42:56 PM
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Rev	Date	Description	Drawn	Appr.
1	20.07.16	ISSUED FOR INFORMATION	EG	BM

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1:1000 @ A1

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Drawn	E. GOODHARDT
Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: ULTIMATE INTERSECTION DESIGN
BALLARTO RD / WOODLANDS RD INTERSECTION
CONCEPT LAYOUT

Status: **PRELIMINARY**
NOT TO BE USED FOR CONSTRUCTION PURPOSES

Date	Scale	Size
20.07.16	1:1000	A1

Drawing Number: V160589-TR-DG-2532
Revision: 1



BY: Userell.goodhardt
DATE PLOTTED: 20/07/2016 4:44:37 PM
CAD File: M:\2016\0501-1000\V160589T_Brompton_Lodge_PSP_Panel\Traffic\Drafting\04 Drawings\01 CAD\V160589-TR-DG-2533.dgn

Rev	Date	Description	Drawn	Appr.
1	20.07.16	ISSUED FOR INFORMATION	EG	BM

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0 10 20 40
1:1000 @ A1

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- FRANKSTON-CRANBOURNE RD - SPEED ZONE 80KM/HR
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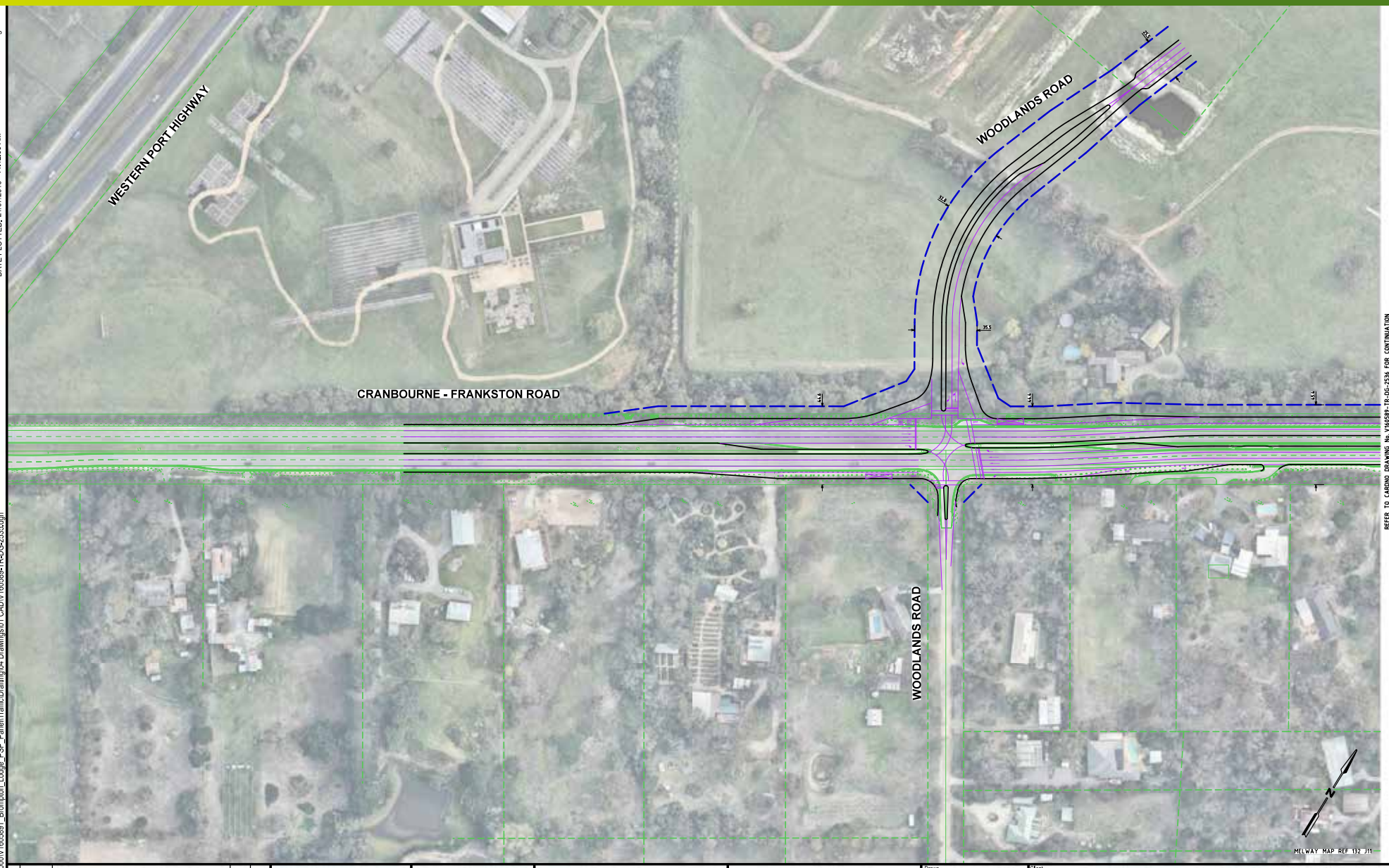
Drawn	E. GOODHARDT
Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: ULTIMATE INTERSECTION DESIGN
BALLARTO RD / EASTERN CONNECTOR RD
CONCEPT LAYOUT

Status: **PRELIMINARY**
NOT TO BE USED FOR CONSTRUCTION PURPOSES

Date	Scale	Size
20.07.16	1:1000	A1

Drawing Number: V160589-TR-DG-2533
Revision: 1



BY: Userell.goodhardt
DATE PLOTTED: 21/07/2016 11:12:00 AM
CAD File:M:\2016\0501-1000\V160589T_Brompton_Lodge_PSP_Panel\Traffic\Drafting\04 Drawings\01 CAD\V160589-TR-DG-2535.dgn

Rev	Date	Description	Drawn	Appr.
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1:1000 @ A1

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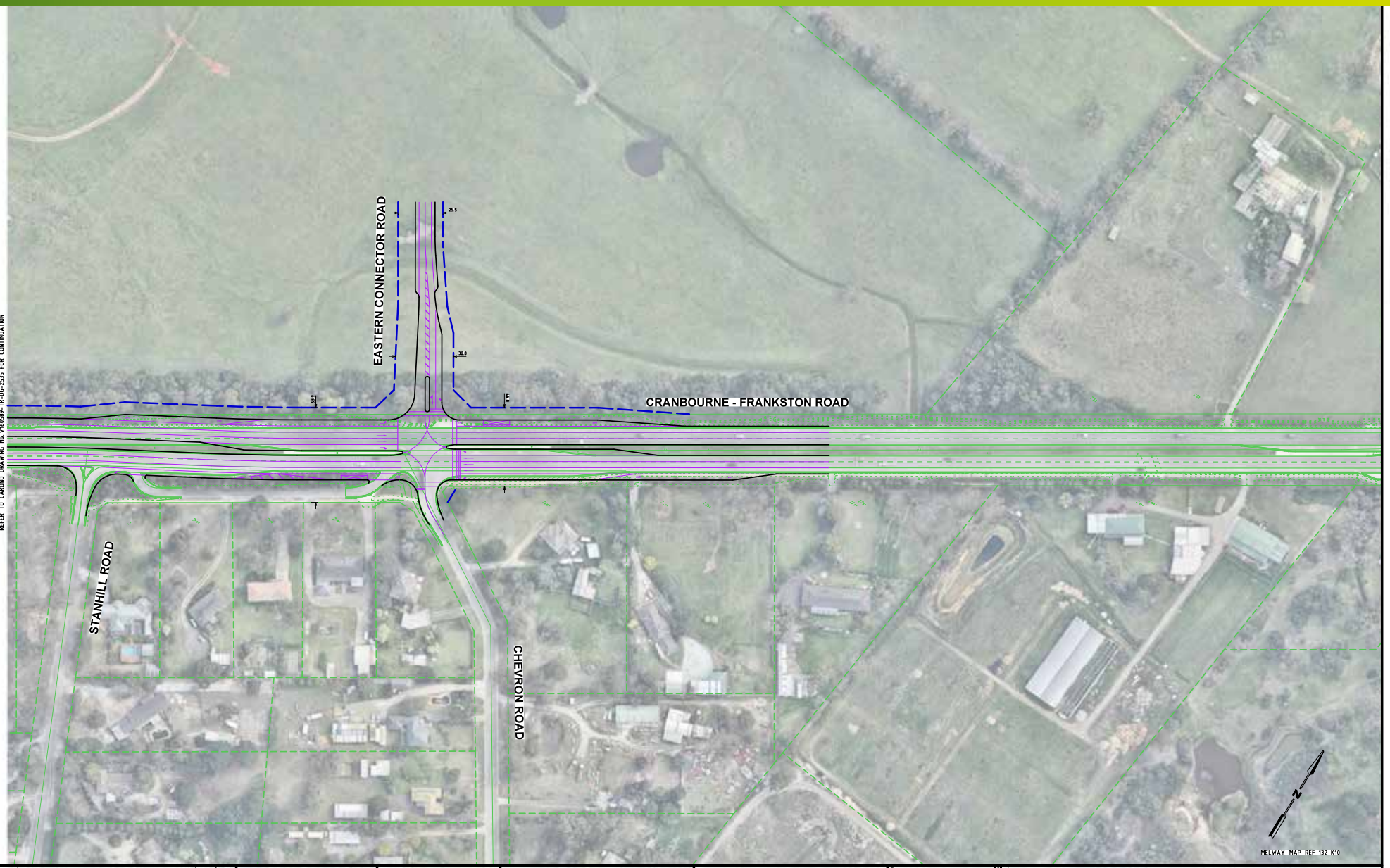
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Drawn	E. GOODHARDT
Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: ULTIMATE INTERSECTION DESIGN
CARNBOURNE - FRANKSTON RD / WOODLANDS RD
CONCEPT LAYOUT

Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date: 20.07.16
Scale: 1:1000
Size: A1
Drawing Number: V160589-TR-DG-2535
Revision: 1



CAD File: M:\2016\0501-1000\V160589T_Brompton_Lodge_PSP_Panel\Traffic\Drafting\04 Drawings\01 CAD\V160589-TR-DG-2536.dgn
DATE PLOTTED: 21/07/2016 11:12:32 AM
BY: User:eli.goodhardt

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1	20.07.16	ISSUED FOR INFORMATION	EG	BM

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1:1000 @ A1

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Drawn	E. GOODHARDT
Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Status: **PRELIMINARY**
NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date: 20.07.16 | Scale: 1:1000 | Size: A1
Title: ULTIMATE INTERSECTION DESIGN
CRANBOURNE - FRANKSTON RD / EASTERN CONNECTOR RD
CONCEPT LAYOUT
Drawing Number: V160589-TR-DG-2536
Revision: 1

APPENDIX F Ultimate Ballarto Road alignment design

BROMPTON LODGE PSP
ULTIMATE ALIGNMENT DESIGN



DRAWING LIST

V160589-TR-DG-1101	COVER SHEET
V160589-TR-DG-2501	BALLARTO ROAD / WESTERN PORT HIGHWAY
V160589-TR-DG-2502	BALLARTO ROAD / WOODLANDS ROAD AND BALLARTO ROAD / EASTERN CONNECTOR ROAD
V160589-TR-DG-2503	BALLARTO ROAD / CRANBOURNE - FRANKSTON ROAD / PEARCEDALE ROAD

N

MELWAY MAP REF 132 & 133

Rev	Date	Description	Drawn	Appr.
2	20.07.16	MPA COMMENTS	EG	BM
1	24.06.16	ISSUED FOR INFORMATION	SGM	BM

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1:5000 @ A1

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Drawn	S. McKENZIE
Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

Client	METROPOLITAN PLANNING AUTHORITY
Project	BROMPTON LODGE PSP CRANBOURNE SOUTH CITY OF CASEY
Title	COVER SHEET - LOCALITY PLAN ULTIMATE ALIGNMENT DESIGN CONCEPT LAYOUT
Status	PRELIMINARY NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date	24.06.16
Scale	1:5000
Size	A1
Drawing Number	V160589-TR-DG-1101
Revision	2

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BY: Usercallgoodhart
DATE PLOTTED: 20/07/2016 2:30:40 PM
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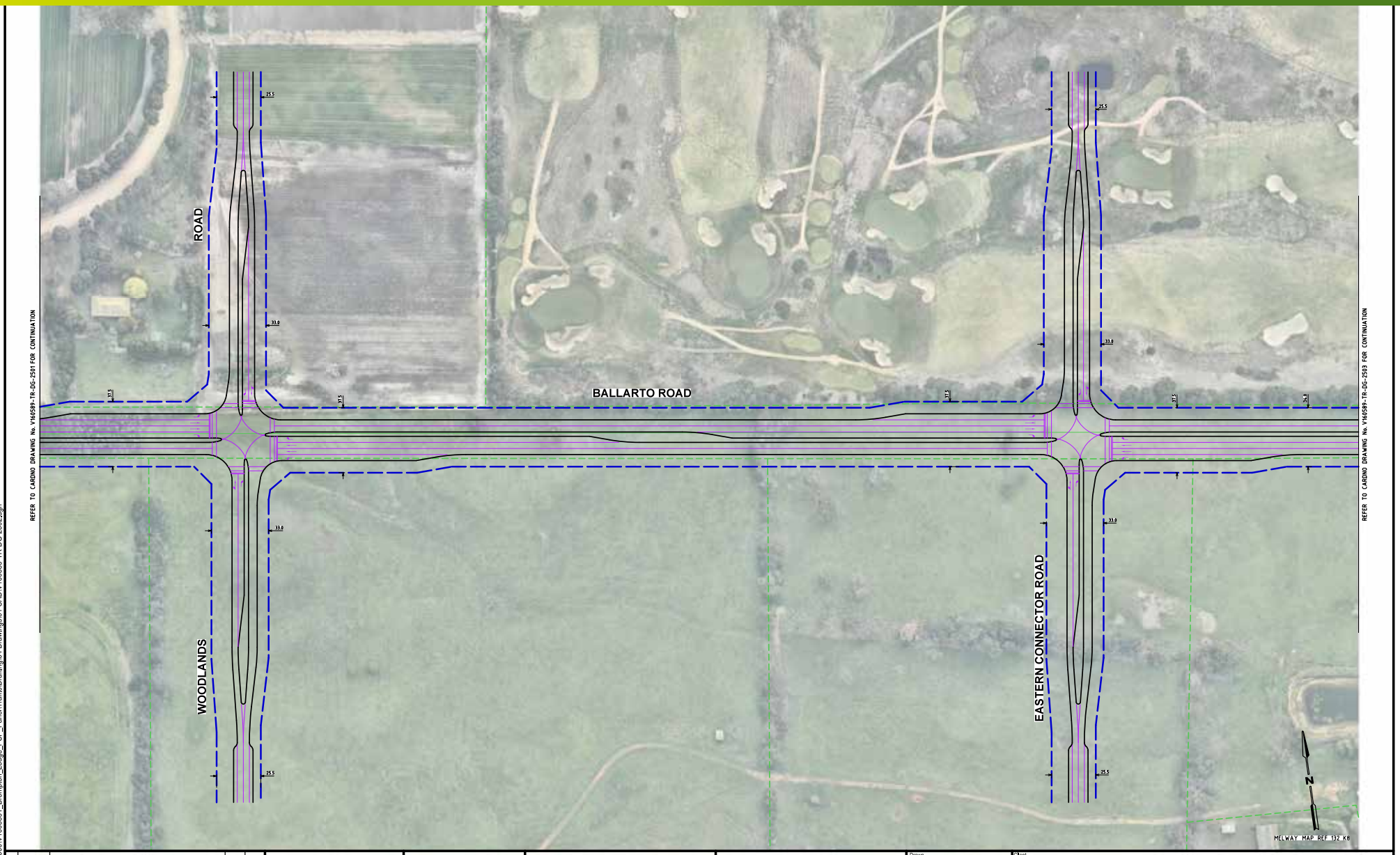
Drawn	S. McKENZIE
Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: ULTIMATE ALIGNMENT DESIGN
WESTERN PORT HIGHWAY / BALLARTO ROAD
CONCEPT LAYOUT

Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES

Date	Scale	Size
24.06.16	1:1000	A1

Drawing Number: V160589-TR-DG-2501
Revision: 2



Rev	Date	Description	Drawn	Appr.
2	20.07.16	MPA COMMENTS	EG	BM
1	24.06.16	ISSUED FOR INFORMATION	SGM	BM

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Designed	S. McKENZIE
Checked	B. HARWOOD
Authorised	B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: ULTIMATE ALIGNMENT DESIGN
BALLARTO RD / WOODLANDS / EASTERN CONNECTOR
CONCEPT LAYOUT

Status: **PRELIMINARY**
NOT TO BE USED FOR CONSTRUCTION PURPOSES

Date	Scale	Size
24.06.16	1:1000	A1

Drawing Number: V160589-TR-DG-2502
Revision: 2

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DATE PLOTTED: 20/07/2016 2:31:43 PM
BY: Usercell.goodhan



Rev	Date	Description	Drawn	Appr.
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2	20.07.16	MPA COMMENTS	EG	BM
1	24.06.16	ISSUED FOR INFORMATION	SGM	BM

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Drawn: S. McKENZIE
Designed: S. McKENZIE
Checked: B. HARWOOD
Authorised: B. MENTHA

Client: METROPOLITAN PLANNING AUTHORITY
Project: BROMPTON LODGE PSP
CRANBOURNE SOUTH
CITY OF CASEY
Title: ULTIMATE ALIGNMENT DESIGN
BALLARTO ROAD / CRANBOURNE - FRANKSTON ROAD
CONCEPT LAYOUT

Status: PRELIMINARY
NOT TO BE USED FOR CONSTRUCTION PURPOSES
Date: 24.06.16
Scale: 1:1000
Size: A1
Drawing Number: V160589-TR-DG-2503
Revision: 3

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DATE PLOTTED: 25/07/2016 12:40:19 PM
BY: Usersellgoodhart

Summary

Cardno

Cost Estimate for Interim Intersection Designs
Summary Sheet
V160589T
Brompton Lodge - Ballarto Road Precinct
Ver 6 - FINAL

21 July 2016

			(Watson's Costing)						
		Ballarto Rd - b/n Westernport Hwy & Woodlands Rd (383m) + b/n Woodlands Rd & Cranbourne-Frankston Rd (880m)	Ballarto Rd / Woodlands Roundabout (Inc. pedestrian crossing)	Cranbourne-Frankston Rd / Woodlands Rd	Cranbourne-Frankston Rd / Eastern Connector Rd	Ballarto Rd / Cranbourne-Frankston Roundabout	Westernport Highway Roundabout connection with Ballarto Rd (to end of island 50m)	
ITEM	DESCRIPTION	RD-01	IN-01	IN-03	IN-04	IN-05	IN-06	TOTAL
A	Project and Program Management	$ 171,688.40	$ 60,618.61	$ 149,232.53	$ 105,189.78	$ 36,569.18	$ 26,674.63	$ 549,973.14
B	Design and Investigation	$ 343,376.80	$ 121,237.23	$ 298,465.07	$ 210,379.57	$ 73,138.37	$ 53,349.25	$ 1,099,946.28
C	Land Acquisition	$ -	$ -	$ -	$ -	$ -	$ -	$ -
D	Construction including Final Design	$ 4,002,060.66	$ 1,306,246.26	$ 2,984,650.65	$ 2,103,795.65	$ 731,383.68	$ 476,758.30	$ 11,604,895.19
	TOTAL A - D	$ 4,517,125.86	$ 1,488,102.10	$ 3,432,348.25	$ 2,419,365.00	$ 841,091.23	$ 556,782.18	$ 13,254,814.62
E	CONTINGENCY							
	Lower Bound Contingency (0% of D)	$ -	$ -	$ -	$ -	$ -	$ -	$ -
	Upper Bound Contingency (20% of D)	$ 800,412.13	$ -	$ 596,930.13	$ 420,759.13	$ 146,276.74	$ 95,351.66	$ 2,059,729.79
F	PROJECT BUDGET							
	Total Lower Bound Estimate	$ 4,517,125.86	$ -	$ 3,432,348.25	$ 2,419,365.00	$ 841,091.23	$ 556,782.18	$ 11,766,712.52
	Total Upper Bound Estimate	$ 5,317,538.00	$ -	$ 4,029,278.38	$ 2,840,124.13	$ 987,367.96	$ 652,133.84	$ 13,826,442.30
G	Project Budget (75% confidence)	$ 5,117,434.96		$ 3,880,045.85	$ 2,734,934.35	$ 950,798.78	$ 628,295.93	$ 13,311,509.86
	Project Budget (Watson's Costing)		$ 1,488,102.10					

AUTHOR: Greg Pollard *Date:* 21/07/2016
REVIEWER: Rob Henry *Date:* 21/07/2016

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 8.58%
RD-01
Ballarto Rd - b/n Westernport Hwy & Woodlands Rd (383m) + b/n Woodlands Rd & Cranbourne-Frankston Rd (880m)

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
A	**Project and Program Management**					**$ 171,688.40**
	Project Management		3.00%	2.57%	$ 103,013	
	Project scoping, planning, scheduling, monitoring, reporting and commissioning				$ -	
	Management of preconstruction activities, development of tender documents and management of contracts				$ -	
	Managing external interfaces, including community liaison, environmental issues, traffic issues etc.				$ -	
	Records management				$ -	
	Program Administration		2.00%	1.72%	$ 68,675	
B	**Design and Investigation**					**$ 343,376.80**
	Traffic Investigations		0.85%	0.73%	$ 29,187	
	Cadastral and Engineering Survey		1.20%	1.03%	$ 41,205	
	Geotechnical investigations Pavement Investigations and Design		1.20%	1.03%	$ 41,205	
	Environmental Investigations			0.00%	$ -	
	Landscape Design			0.00%	$ -	
	Preliminary and Final Design		4.25%	3.65%	$ 145,935	
	Construction Management		2.50%	2.15%	$ 85,844	
C	**Land Acquisition**					**$ -**
	Acquire land				$ -	
D	**Construction**				$ 4,002,061	**$ 4,002,060.66**
1	**PRELIMINARIES**					$ 120,000
1.1	Site Establishment	1	item	$ 35,000.00	$ 35,000	
1.2	Site Management & supervision including QA	26	wks.	$ 2,500.00	$ 65,000	
1.3	Provision for Traffic Management	1	item	$ 20,000.00	$ 20,000	
2	**DEMOLITION**					$ 74,300
2.1	Trees	1	Item	$ 65,000.00	$ 65,000	
2.2	Existing Fencing	1	Item	$ 6,000.00	$ 6,000	
2.3	Existing K&C	66	m	$ 50.00	$ 3,300	
3	**EARTHWORKS**					$ 352,391
3.1	Stripping topsoil (150mm)	17053	m²	$ 6.00	$ 102,317	
3.2	Excavation and removal	5557	m³	$ 30.00	$ 166,716	
3.3		5557	m²	$ 15.00	$ 83,358	
4	**PAVEMENT**					$ 1,561,068
4.1	Deep lift asphalt 195 mm	8841	m²	$ 120.00	$ 1,060,920	
4.2	Subbase course - 250 mm 3% CTCR	11114	m²	$ 45.00	$ 500,148	
5	**DRAINAGE**					$ 623,359
5.1	subsoil drains 100mm dia - screenings	2526	m	$ 45.00	$ 113,670	
5.2	subsoil drains 100mm dia - no fines conc	14	m	$ 50.00	$ 700	
5.3	375 RCP (Class 2)	598	m	$ 200.00	$ 119,564	
5.4	450 RCP (Class 2)	421	m	$ 275.00	$ 115,775	
5.5	525 RCP (Class 2)	421	m	$ 350.00	$ 147,350	
5.6	Drainage Pits	51	No	$ 2,500.00	$ 126,300	
6	**CONCRETE WORKS**					$ 884,100
6.1	Kerb &channel	2526	m	$ 50.00	$ 126,300	
6.2	3m wide shared footpath (125mm double reo 120mm FCR	7578	m²	$ 100.00	$ 757,800	
7	**LANDSCAPING WORKS**					$ 76,500
7.1	Topsoiling seeding	9000	m²	$ 8.50	$ 76,500	
8	**SIGNING**	1	item	$ 10,000	$ 10,000	$ 10,000
9	**LINEMARKING (Thermoplastic)**	1	item	$ 30,000	$ 30,000	$ 30,000
10	**SERVICE RELOCATION**					$ -
10.1	(Water, Gas, Telstra) NOT INCLUDED IN EST		not included		$ -	
11	**POWER & LIGHTING /Relocation**					$ 270,343
11.1	Light pole (1-way) and luminous lights	18	No	$ 8,000.00	$ 144,343	
11.3	Cabling, conduits and pits	1	item	$ 100,000.00	$ 100,000	
11.4	Electrical connection	1	Item	$ 26,000.00	$ 26,000	
12	**MISCELLANEOUS**					$ -
13	**PROVISIONAL SUM - DAYWORK**		item		$ -	$ -
	TOTAL A - D				$ 4,517,126	**$ 4,517,125.86**
E	**Contingency**					
	Lower Bound Contingency (0% of D)				0%	$ -

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 8.58%
RD-01
Ballarto Rd - b/n Westernport Hwy & Woodlands Rd (383m) + b/n Woodlands Rd & Cranbourne-Frankston Rd (880m)

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
	Upper Bound Contingency (20% of D)				20%	$ 800,412.13
F	PROJECT BUDGET					
	Lower Bound Estimate					$ 4,517,125.86
	Upper Bound Estimate					$ 5,317,538.00
G	Project Budget (75% Confidence)					$ 5,117,434.96

Comments Linear metre rate $ 4,052

BROMPTON LODGE PRECINCT
INTERSECTION COSTS
IN-01 (Woodlands Rd / Ballarto) **- Interim Alignment Design - Option 2, Woodlands Drive Roundabout Concept Layout (DWG Ref: V160589 - TR - DG - 2521, Rev 1)**
Watsons Estimate_20160429

Item	Actual Qty	Unit	Rate	Actual amount	Sub Total	Comments
PROJECT AND PROGRAM MANAGEMENT					$ 61,172.75	Percentages as per Cardno estimate
PROJECT MANAGEMENT		2.81%		$ 36,651.48		
Project scoping, planning, scheduling,						
Management and preconstruction activities,						
Managing external interfaces, including						
Records Management						
Program administration		1.88%		$ 24,521.27		
DESIGN AND INVESTIGATION					$ 122,606.36	Percentages as per Cardno estimate
Traffic Investigations		0.80%		$ 10,434.58		
Cadastral and Engineering Survey		1.13%		$ 14,738.85		
Geotechnical investigations, Pavement		1.13%		$ 14,738.85		
Environmental investigations						
Landscape Design						
Preliminary and Final Design		3.99%		$ 52,042.49		
Construction Management		2.35%		$ 30,651.59		
LAND ACQUISITION						Not included
Aquire Land						
CONSTRUCTION (includes 3% rate rise)					**$ 1,304,322.99**	
PRELIMINARIES					$ 80,000.00	
Site Establishment	1	item	$ 35,000	$ 35,000.00		
Site Management & supervision including QA	16	weeks	$ 2,500	$ 40,000.00		
Provision for Traffic	1	item	$ 5,000	$ 5,000.00		No traffic management required. Greenfields
DEMOLITION					$ 30,000.00	
Trees	1	Item	$ 30,000	$ 30,000.00		
Existing Fence						
EARTHWORKS					$ 124,414.25	
Stripping topsoil (150mm)	8500	sqm	$ 1.45	$ 12,325.00		
Excavation and removal (including reclamation	2653	cum	$ 27.25	$ 72,294.25		
Soft Spot Rectification (reclaimed FCR)	2653	sqm	$ 15	$ 39,795.00		
PAVEMENT					$ 483,640.00	
Deep Lift Asphalt (195mm)	2440	sqm	$ 115	$ 280,600.00		
Subbase Course - 250mm 3% CTCR	3384	sqm	$ 60	$ 203,040.00		
DRAINAGE					$ 141,661.25	
Subsoil Drains 100mm dia - scrrenings	745	m	$ 26	$ 19,556		
Subsoil Drains 100mm dia - no fines concrete	0	m	$ 39	$ -		
Flushout Risers	1	item	$ 10,000	$ 10,000		
375 RCP	320	m	$ 224	$ 71,680		
Drainage pit	13	No	$ 2,725	$ 35,425		
CCTV	1	item	$ 5,000	$ 5,000		
CONCRETE WORKS					$ 160,036.00	
Kerb and Channel	745	m	$ 47	$ 34,866		
Kerb and Channel (Semi Mountable)						
Trafficable Pavement						
1.5m wide footpath	0	sqm	$ 50	$ -		
3m shared path (trafficable)	1284	sqm	$ 65	$ 83,460		
Concrete Island infill	520	sqm	$ 65	$ 33,800		
Pram Crossings	14	No	$ 565	$ 7,910		
LANDSCAPING WORKS					$ 13,331.50	
Topsoiling Seeding	3809	sqm	$ 4	$ 13,332		
SIGNING	1	item	$ 5,000	$ 5,000	$ 5,000.00	
LINEMARKING (Thermoplastic) inc Bus Lane	1	item	$ 12,000	$ 12,000	$ 12,000.00	
SERVICE RELOCATION					$ -	
O/H Electrical	0	item	$ 150,000	$ -		Assumed no service relocations required
Gas	0	item	$ 60,000	$ -		
Communications	0	item	$ 60,000	$ -		
TRAFFIC SIGNAL					$ 86,250.00	As per Cardno estimate
General Items		item	$ 27,500	$ -		
Conduits		item	$ 28,000	$ -		
Pedestals	1	item	$ 86,250	$ 86,250		
Lanterns		item	$ 25,000	$ -		
Controller		item	$ 82,500	$ -		
Detectors		item	$ 15,000	$ -		
Cabling & connections		item	$ 48,000	$ -		
Clean Up		item	$ 11,000	$ -		
AUTHORITY						
POWER & LIGHTING					$ 127,000.00	As per Cardno estimate
JUP Lighting Single		No	$ 4,000	$ -		
New Light Pole - single		No	$ 8,000	$ -		
New Light Pole - double		No	$ 6,000	$ -		
Conduits & Pits		item	$ 25,000	$ -		
Electrical Connection	1	item	$ 127,000	$ 127,000		
MISCELLANEOUS					$ 3,000.00	
Coordinate with Service Authorities	1	item	$ 1,500	$ 1,500		
Permanent Survey Marks	1	item	$ 1,500	$ 1,500		
PROVISIONAL SUM - DAYWORK						
TOTAL				$ 1,450,112.11	**$ 1,488,102.10**	
Costing excludes land component for this Project set out in Table 7 of the Exhibited Brompton Lodge DCP for IN-01						
0.64Ha - $428,887.00						

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 10.00%
IN-03
Cranbourne-Frankston Rd / Woodlands Rd

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
A	**Project and Program Management**					**$ 149,233**
	Project Management		3.00%	3.00%	$ 89,540	
	Project scoping, planning, scheduling, monitoring, reporting and commissioning				$ -	
	Management of preconstruction activities, development of tender documents and management of contracts				$ -	
	Managing external interfaces, including community liaison, environmental issues, traffic issues etc				$ -	
	Records management				$ -	
	Program Administration		2.00%	2.00%	$ 59,693	
B	**Design and Investigation**					**$ 298,465**
	Traffic Investigations		0.85%	0.85%	$ 25,370	
	Cadastral and Engineering Survey		1.20%	1.20%	$ 35,816	
	Geotechnical investigations Pavement Investigations and Design		1.20%	1.20%	$ 35,816	
	Environmental Investigations			0.00%	$ -	
	Landscape Design			0.00%	$ -	
	Preliminary and Final Design		4.25%	4.25%	$ 126,848	
	Construction Management		2.50%	2.50%	$ 74,616	
C	**Land Acquisition**					**$ -**
	Acquire land				$ -	
D	**Construction**				$ 2,984,651	**$ 2,984,651**
1	**PRELIMINARIES**					$ 173,000
1.1	Site Establishment	1	item	$ 68,000.00	$ 68,000	
1.2	Site Management & supervision including QA	18	wks	$ 2,500.00	$ 45,000	
1.3	Provision for traffic	1	item	$ 60,000.00	$ 60,000	
2	**DEMOLITION**					$ 64,475
2.1	Kerb and channel	512	m	$ 45.00	$ 23,040	
2.2	Existing granatic sand footpath	75	m	$ 6.00	$ 450	
2.3	Trees	12	No	$ 1,000.00	$ 12,000	
2.4	Steel guard rails	80	m	$ 10.00	$ 800	
2.5	High tension wire guard rails	423	m	$ 15.00	$ 6,345	
2.6	Concrete island	268	m²	$ 65.00	$ 17,420	
2.7		68	m²	$ 65.00	$ 4,420	
3	**EARTHWORKS**					$ 189,390
3.1	Stripping topsoil (150mm)	12253	m²	$ 6.00	$ 73,517	
3.2	Excavation and removal (inc. reclamation of FCR)	2575	m³	$ 30.00	$ 77,249	
3.3	Soft Spot Rectification (reclaimed FCR)	2575	m²	$ 15.00	$ 38,624	
4	**PAVEMENT**					$ 892,616
4.1	Deep lift asphalt 195 mm	4501	m²	$ 120.00	$ 540,120	
4.2	Subbase course - 250 mm 3% CTCR	5150	m²	$ 45.00	$ 231,746	
4.3	Asphalt resheet	3450	m²	$ 35.00	$ 120,750	
5	**DRAINAGE**					$ 326,095
5.1	subsoil drains 100mm dia - screenings	1041	m	$ 45.00	$ 46,845	
5.2	subsoil drains 100mm dia - no fines conc	380	m	$ 50.00	$ 19,000	
5.3	375 RCP (Class 2)	1041	m	$ 200.00	$ 208,200	
5.4	drainage pit	18	No	$ 2,500.00	$ 44,550	
5.5	modify existing drainage pit	3	No	$ 2,500.00	$ 7,500	
6	**CONCRETE WORKS**					$ 452,965
6.1	Kerb &channel	1442	m	$ 60.00	$ 86,520	
6.2	1.5m wide footpath	233	m	$ 65.00	$ 15,145	
6.3	3m shared path (trafficable)	2760	m²	$ 100.00	$ 276,000	
6.4	Concrete island infill	683	m²	$ 100.00	$ 68,300	
6.5	Bus shelter slab	2	No	$ 3,500.00	$ 7,000	
7	**LANDSCAPING WORKS**					$ 31,110
7.1	Topsoiling seeding	3660	m²	$ 8.50	$ 31,110.00	
8	**SIGNING**	1	item	$ 7,000	$ 7,000	$ 7,000
9	**LINEMARKING (Thermoplastic) inc Bus lane treatment**	1	item	$ 25,000	$ 25,000	$ 25,000
10	**SERVICE RELOCATION**					$ 270,000
10.1	Overhead electrical	1	item	$ 150,000.00	$ 150,000	
10.2	Gas	1	item	$ 60,000.00	$ 60,000	
10.3	Water	1	item	$ 60,000.00	$ 60,000	
11	**TRAFFIC SIGNAL**					$ 402,000
11.1	10 year VicRoads maintenance	1	Item	$ 100,000	$ 100,000	

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 10.00%
IN-03
Cranbourne-Frankston Rd / Woodlands Rd

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
11.2	General items	1	Item	$ 27,500	$ 27,500	
11.3	Conduits	1	Item	$ 28,000	$ 28,000	
11.4	Pedestals	1	Item	$ 65,000	$ 65,000	
11.5	Lanterns	1	Item	$ 25,000	$ 25,000	
11.6	Controller	1	Item	$ 82,500	$ 82,500	
11.7	Detectors	1	Item	$ 15,000	$ 15,000	
11.8	Cabling & connections	1	Item	$ 48,000	$ 48,000	
11.9	Clean-up	1	Item	$ 11,000	$ 11,000	
12	**POWER & LIGHTING /Relocation**					$ 151,000
12.1	O/H Powerlines Relocation	1	No	$ 8,000.00	$ 8,000	
12.2	power pole relocation (HV & LV)	1	No	$ 12,000.00	$ 12,000	
11.3	JUP lighting single	2	No	$ 4,000	$ 8,000	
12.4	New light pole - single	7	No	$ 8,000.00	$ 56,000	
12.5	New light pole - double	2	No	$ 6,000.00	$ 12,000	
12.6	conduits and pits	1	item	$ 25,000.00	$ 25,000	
12.7	Electrical connection	1	item	$ 30,000.00	$ 30,000	
13	**MISCELLANEOUS**					$ -
13.1	Driveway crossings		not included		$ -	
13,2	Conduits		not included		$ -	
14	**PROVISIONAL SUM - DAYWORK**		item		$ -	$ -
	TOTAL A - D				$ 3,432,348	**$ 3,432,348**
E	**Contingency**					
	Upper Bound Contingency (20% of D)				0%	$ -
	0				20%	$ 596,930.13
F	**PROJECT BUDGET**					
	Lower Bound Estimate					$ 3,432,348
	Upper Bound Estimate					$ 4,029,278
G	**Project Budget (75% Confidence)**					**$ 3,880,046**

Comments
NB Estimate does not include reinstatement of high tension wire guard rail

IN-04

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 10.00%
IN-04
Cranbourne-Frankston Rd / Eastern Connector Rd

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
A	**Project and Program Management**					**$ 105,190**
	Project Management		3.00%	3.00%	$ 63,114	
	Project scoping, planning, scheduling, monitoring, reporting and commissioning				$ -	
	Management of preconstruction activities, development of tender documents and management of contracts				$ -	
	Managing external interfaces, including community liaison, environmental issues, traffic issues etc				$ -	
	Records management				$ -	
	Program Administration		2.00%	2.00%	$ 42,076	
B	**Design and Investigation**					**$ 210,380**
	Traffic Investigations		0.85%	0.85%	$ 17,882	
	Cadastral and Engineering Survey		1.20%	1.20%	$ 25,246	
	Geotechnical investigations Pavement Investigations and Design		1.20%	1.20%	$ 25,246	
	Environmental Investigations			0.00%	$ -	
	Landscape Design			0.00%	$ -	
	Preliminary and Final Design		4.25%	4.25%	$ 89,411	
	Construction Management		2.50%	2.50%	$ 52,595	
C	**Land Acquisition**					**$ -**
	Acquire land				$ -	
D	**Construction**				$ 2,103,796	**$ 2,103,796**
1	**PRELIMINARIES**					$ 145,000
1.1	Site Establishment	1	item	$ 45,000.00	$ 45,000	
1.2	Site Management & supervision including QA	16	wks	$ 2,500.00	$ 40,000	
1.3	Provision for traffic	1	item	$ 60,000.00	$ 60,000	
2	**DEMOLITION**					$ 73,470
2.1	Kerb and channel	512	m	$ 45.00	$ 23,040	
2.3	Trees	15	No	$ 1,000.00	$ 15,000	
2.4	Concrete guard rail	1	Item	$ 10,000.00	$ 10,000	
2.5	High tension wire guard rails	391	m	$ 15.00	$ 5,865	
2.6	Concrete island	216	m²	$ 65.00	$ 14,040	
2.7	Redundant concrete bus bay	85	m²	$ 65.00	$ 5,525	
3						$ 93,734
3.1	Stripping topsoil (150mm)	7414	m²	$ 6.00	$ 44,483	
3.2	Excavation and removal (inc. reclamation of FCR)	1094	m³	$ 30.00	$ 32,834	
3.3	Soft Spot Rectification (reclaimed FCR)	1094	m²	$ 15.00	$ 16,417	
4	**PAVEMENT**					$ 471,879
4.1	Deep lift asphalt 195 mm	1630	m²	$ 135.00	$ 220,050	
4.2	Subbase course - 250 mm 3% CTCR	2189	m²	$ 65.00	$ 142,279	
4.3	Asphalt resheet	3130	m²	$ 35.00	$ 109,550	
5	**DRAINAGE**					$ 151,360
5.1	subsoil drains 100mm dia - screenings	448	m	$ 45.00	$ 20,160	
5.2	subsoil drains 100mm dia - no fines conc	384	m	$ 50.00	$ 19,200	
5.3	375 RCP (Class 2)	448	m	$ 200.00	$ 89,600	
5.4	drainage pit	6	No	$ 2,500.00	$ 14,900	
5.5	modify existing drainage pit	3	No	$ 2,500.00	$ 7,500	
6	**CONCRETE WORKS**					$ 264,850
6.1	Kerb &channel	1242	m	$ 60.00	$ 74,520	
6.2	1.5m wide footpath	122	m	$ 65.00	$ 7,930	
6.3	3m shared path (trafficable)	1707	m²	$ 100.00	$ 170,700	
6.4	Concrete island infill	47	m²	$ 100.00	$ 4,700	
6.5	Bus shelter slab	2	No	$ 3,500.00	$ 7,000	
7	**LANDSCAPING WORKS**					$ 29,504
7.1	Topsoiling seeding	3471	m²	$ 8.50	$ 29,503.50	
8	**SIGNING**	1	item	$ 7,000	$ 7,000	$ 7,000
9	**LINEMARKING (Thermoplastic) inc Bus lane treatment**	1	item	$ 20,000	$ 20,000	$ 20,000
10	**SERVICE RELOCATION**					$ 310,000
10.1	Overhead electrical	1	item	$ 100,000.00	$ 100,000	
10.2	Gas	1	item	$ 150,000.00	$ 150,000	
10.3	Telecom	1	item	$ 60,000.00	$ 60,000	
11	**TRAFFIC SIGNAL**		item			$ 402,000
11.1	10 year VicRoads maintenance	1	Item	$ 100,000	$ 100,000	
11.2	General items	1	Item	$ 27,500	$ 27,500	

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 10.00%
IN-04
Cranbourne-Frankston Rd / Eastern Connector Rd

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
11.3	Conduits	1	Item	$ 28,000	$ 28,000	
11.4	Pedestals	1	Item	$ 65,000	$ 65,000	
11.5	Lanterns	1	Item	$ 25,000	$ 25,000	
11.6	Controller	1	Item	$ 82,500	$ 82,500	
11.7	Detectors	1	Item	$ 15,000	$ 15,000	
11.8	Cabling & connections	1	Item	$ 48,000	$ 48,000	
11.9	Clean-up	1	Item	$ 11,000	$ 11,000	
12	**POWER & LIGHTING /Relocation**					$ 135,000
12.1	O/H Powerlines Relocation	1	No	$ 8,000.00	$ 8,000	
12.2	power pole relocation (HV & LV)	1	No	$ 12,000.00	$ 12,000	
12.3	JUP lighting single	2	No	$ 4,000	$ 8,000	
12.4	New light pole - single	5	No	$ 8,000.00	$ 40,000	
12.5	New light pole - double	2	No	$ 6,000.00	$ 12,000	
12.6	conduits and pits	1	item	$ 25,000.00	$ 25,000	
12.7	Electrical connection	1	item	$ 30,000.00	$ 30,000	
13	**MISCELLANEOUS**					$ -
13.1	Driveway crossings		not included		$ -	
13.2	Conduits		not included		$ -	
14	**PROVISIONAL SUM - DAYWORK**		item		$ -	$ -
	TOTAL A - D				$ 2,419,365	**$ 2,419,365**
E	**Contingency**					
	Upper Bound Contingency (20% of D)				0%	$ -
	0				20%	$ 420,759.13
F	**PROJECT BUDGET**					
	Lower Bound Estimate					$ 2,419,365
	Upper Bound Estimate					$ 2,840,124
G	**Project Budget (75% Confidence)**					**$ 2,734,934**

Comments
NB Estimate does not include reinstatement of high tension wire guard rail

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 10.00%
IN-05
Ballarto Rd / Cranbourne-Frankston Roundabout

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
A	**Project and Program Management**					**$ 36,569**
	Project Management		3.00%	3.00%	$ 21,942	
	Project scoping, planning, scheduling, monitoring, reporting and commissioning				$ -	
	Management of preconstruction activities, development of tender documents and management of contracts				$ -	
	Managing external interfaces, including community liaison, environmental issues, traffic issues etc				$ -	
	Records management				$ -	
	Program Administration		2.00%	2.00%	$ 14,628	
B	**Design and Investigation**					**$ 73,138**
	Traffic Investigations		0.85%	0.85%	$ 6,217	
	Cadastral and Engineering Survey		1.20%	1.20%	$ 8,777	
	Geotechnical investigations Pavement Investigations and Design		1.20%	1.20%	$ 8,777	
	Environmental Investigations			0.00%	$ -	
	Landscape Design			0.00%	$ -	
	Preliminary and Final Design		4.25%	4.25%	$ 31,084	
	Construction Management		2.50%	2.50%	$ 18,285	
C	**Land Acquisition**					**$ -**
	Acquire land				$ -	
D	**Construction**				$ 731,384	**$ 731,384**
1	**PRELIMINARIES**					$ 113,000
1.1	Site Establishment	1	item	$ 18,000.00	$ 18,000	
1.2	Site Management & supervision including QA	14	wks	$ 2,500.00	$ 35,000	
1.3	Provision for traffic	1	item	$ 60,000.00	$ 60,000	
2	**DEMOLITION**					$ 12,000
2.1	Trees	12	No	$ 1,000.00	$ 12,000	
3	**EARTHWORKS**					$ 50,529
3.1	Stripping topsoil (150mm)	3105	m²	$ 6.00	$ 18,630	
3.2	Excavation and removal (inc. reclamation of FCR)	576	m³	$ 30.00	$ 17,266	
3.3	Asphalt planing	1	item	$ 6,000.00	$ 6,000	
3.4	Soft Spot Rectification (reclaimed FCR)	576	m²	$ 15.00	$ 8,633	
4						$ 216,483
4.1	Deep lift asphalt 195 mm	1039	m²	$ 135.00	$ 140,265	
4.2	Subbase course - 250 mm 3% CTCR	1151	m²	$ 65.00	$ 74,818	
4.3	Asphalt resheet	40	m²	$ 35.00	$ 1,400	
5	**DRAINAGE**					$ 59,285
5.1	subsoil drains 100mm dia - screenings	193	m	$ 45.00	$ 8,685	
5.2	subsoil drains 100mm dia - no fines conc	40	m	$ 50.00	$ 2,000	
5.3	375 RCP (Class 2)	193	m	$ 200.00	$ 38,600	
5.4	drainage pit	4	No	$ 2,500.00	$ 10,000	
6	**CONCRETE WORKS**					$ 117,440
6.1	Kerb &channel	249	m	$ 60.00	$ 14,940	
6.2	1.5m wide footpath	0	m	$ 65.00	$ -	
6.3	3m shared path (trafficable)	915	m²	$ 100.00	$ 91,500	
6.4	Concrete island infill	110	m²	$ 100.00	$ 11,000	
7	**LANDSCAPING WORKS**					$ 7,897
7.1	Topsoiling seeding	929	m²	$ 8.50	$ 7,896.50	
8	**SIGNING**	1	item	$ 3,000	$ 3,000	$ 3,000
9	**LINEMARKING (Thermoplastic)**	1	item	$ 15,000	$ 15,000	$ 15,000
10	**SERVICE RELOCATION**					$ -
10.1	(Water, Gas, Telstra) NOT INCLUDED IN EST		not included		$ -	
10.2	hydrant relocated		item		$ -	
11	**PEDESTRIAN OPERATED TRAFFIC SIGNAL**		item			$ 71,750
11.1	General items	1	Item	$ 6,875	$ 6,875	
11.2	Conduits	1	Item	$ 7,000	$ 7,000	
11.3	Pedestals	1	Item	$ 16,250	$ 16,250	
11.4	Lanterns	1	Item	$ 6,250	$ 6,250	
11.5	Controller	1	Item	$ 20,625	$ 20,625	
11.6	Cabling & connections	1	Item	$ 12,000	$ 12,000	
11.7	Clean-up	1	Item	$ 2,750	$ 2,750	
12	**POWER & LIGHTING /Relocation**					$ 65,000
12.1	JUP lighting single	2	No	$ 4,000	$ 8,000	
12.2	New light pole - single	2	No	$ 8,000.00	$ 16,000	

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 10.00%
IN-05
Ballarto Rd / Cranbourne-Frankston Roundabout

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
12.3	New light pole - double	1	No	$ 6,000.00	$ 6,000	
12.4	conduits and pits	1	item	$ 15,000.00	$ 15,000	
12.5	Electrical connection	1	item	$ 20,000.00	$ 20,000	
13	**MISCELLANEOUS**					$ -
13.1	Driveway crossings		not included		$ -	
13.2	Conduits		not included		$ -	
14	**PROVISIONAL SUM - DAYWORK**		item		$ -	$ -
	TOTAL A - D				$ 841,091	**$ 841,091**
E	**Contingency**					
	Upper Bound Contingency (20% of D)				0%	$ -
	0				20%	$ 146,276.74
F	**PROJECT BUDGET**					
	Lower Bound Estimate					$ 841,091
	Upper Bound Estimate					$ 987,368
G	**Project Budget (75% Confidence)**					**$ 950,799**

Comments

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 11.19%
IN-06
Westernport Highway Roundabout connection with Ballarto Rd (to end of island 50m)

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
A	**Project and Program Management**					**$ 26,674.63**
	Project Management		3.00%	3.36%	$ 16,005	
	Project scoping, planning, scheduling, monitoring, reporting and commissioning				$ -	
	Management of preconstruction activities, development of tender documents and management of contracts				$ -	
	Managing external interfaces, including community liaison, environmental issues, traffic issues etc.				$ -	
	Records management				$ -	
	Program Administration		2.00%	2.24%	$ 10,670	
B	**Design and Investigation**					**$ 53,349.25**
	Traffic Investigations		0.85%	0.95%	$ 4,535	
	Cadastral and Engineering Survey		1.20%	1.34%	$ 6,402	
	Geotechnical investigations Pavement Investigations and Design		1.20%	1.34%	$ 6,402	
	Environmental Investigations			0.00%	$ -	
	Landscape Design			0.00%	$ -	
	Preliminary and Final Design		4.25%	4.76%	$ 22,673	
	Construction Management		2.50%	2.80%	$ 13,337	
C	**Land Acquisition**					**$ -**
	Acquire land				$ -	
D	**Construction**				$ 476,758	**$ 476,758.30**
1	**PRELIMINARIES**					$ 102,300
1.1	Site Establishment	1	item	$ 12,300.00	$ 12,300	
1.2	Site Management & supervision including QA	12	wks.	$ 2,500.00	$ 30,000	
1.3	Provision for Traffic Management	1	item	$ 60,000.00	$ 60,000	
2	**DEMOLITION**					$ 11,250
2.1	Trees	1	Item	$ 7,000.00	$ 7,000	
2.2	Existing Fencing	1	Item	$ 2,000.00	$ 2,000	
2.3	Existing K & C	45	Lm	$ 50.00	$ 2,250	
3	**EARTHWORKS**					$ 20,055
3.1	Stripping topsoil (150mm)	1242	m²	$ 6.00	$ 7,451	
3.2	Excavation and removal	280	m³	$ 30.00	$ 8,403	
3.4	Soft Spot Rectification 50%of pavement	280	m²	$ 15.00	$ 4,202	
4	**PAVEMENT**					$ 115,748
4.1	Deep lift asphalt 195 mm	436	m²	$ 135.00	$ 58,860	
4.2	Subbase course - 250 mm 3% CTCR	560	m²	$ 65.00	$ 36,413	
4.3	Asphalt Overlay/Regulation	455	m²	$ 45.00	$ 20,475	
	Saw cut existing pavement provide heavy tack					
5	**DRAINAGE**					$ 32,250
5.1	subsoil drains 100mm dia - screenings	200	m	$ 45.00	$ 9,000	
5.2	subsoil drains 100mm dia - no fines conc	25	m	$ 50.00	$ 1,250	
	Flushout Risers					
5.3	375 RCP (Class 2)	60	m	$ 200.00	$ 12,000	
5.6	Drainage Pits	4	No	$ 2,500.00	$ 10,000	
	CCTV					
	Head Walls					
6	**CONCRETE WORKS**					$ 61,525
6.1	Kerb &channel (Barrier)	90	m	$ 50.00	$ 4,500	
6.2	Kerb &channel (Semi-mountable)	155	m	$ 55.00	$ 8,525	
6.5	3m wide shared footpath (125mm double reo 120mm FCR	300	m	$ 100.00	$ 30,000	
6.6	Concrete island infill	151	m²	$ 100.00	$ 15,100	
6.7	Pram Crossing	4	No.	$ 850.00	$ 3,400	
7	**LANDSCAPING WORKS**					$ 6,630
7.1	Topsoiling seeding	780	m²	$ 8.50	$ 6,630.00	
8	**SIGNING**	1	item	$ 3,000	$ 3,000	$ 3,000
9	**LINEMARKING (Thermoplastic)**	1	item	$ 10,000	$ 10,000	$ 10,000
10	**SERVICE RELOCATION**					$ -
10.1	(Water, Gas, Telstra) NOT INCLUDED IN EST		not included		$ -	
11	**POWER & LIGHTING /Relocation**					$ 114,000
11.1	Light pole (1-way) and luminous lights	5	No	$ 8,000.00	$ 40,000	
11.1	Light pole (2-way) and luminous lights	4	No	$ 8,000.00	$ 32,000	
11.3	Cabling, conduits and pits	1	item	$ 20,000.00	$ 20,000	

Cost Estimate for Interim Intersection Designs **21 July 2016**
V160589T
Brompton Lodge - Ballarto Road Precinct **PM %:** 11.19%
IN-06
Westernport Highway Roundabout connection with Ballarto Rd (to end of island 50m)

ITEM	DESCRIPTION	QUANTITY	UNIT	RATE	AMOUNT	SUBTOTAL
11.4	Electrical connection	1	Item	$ 22,000.00	$ 22,000	
12	**MISCELLANEOUS**					$ -
	Permanent Survey Marks					
13	**PROVISIONAL SUM - DAYWORK**		item		$ -	$ -
	TOTAL A - D				$ 556,782	**$ 556,782.18**
E	**Contingency**					
	Lower Bound Contingency (0% of D)				0%	$ -
	Upper Bound Contingency (20% of D)				20%	$ 95,351.66
F	**PROJECT BUDGET**					
	Lower Bound Estimate					$ 556,782.18
	Upper Bound Estimate					$ 652,133.84
G	**Project Budget (75% Confidence)**					**$ 628,295.93**

Comments
NB No allowance for service's alteraton in Westernport Hwy reserve


METROPOLITAN
PLANNING
AUTHORITY